



European University Alliance for Global Health

eduTAP

<https://eduTAP.eu>

A Concept and System of Digital Identities for proximity on-site service access

Modern Campus ID (Student ID / Staff ID) for universities in Europe & the world

Alexander Loechel & Simon Lund - EUNIS 2025 - Belfast



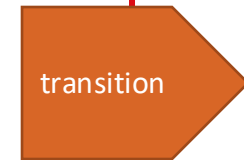
The LMU Student ID (over time)

LMU Student ID till 2019

LMUcard Student ID 2019-2025

LMU Student ID 2025+

LMU Student ID 2027+ ?

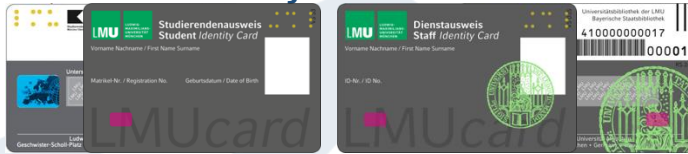


today

eduTAP & eduTAP@LMU

The Way to eduTAP

- LMUcard Project 2015-2019



- The European Student Card



European
Student Card

- The ECCA Conference 2019 Linz



- EUGLOH

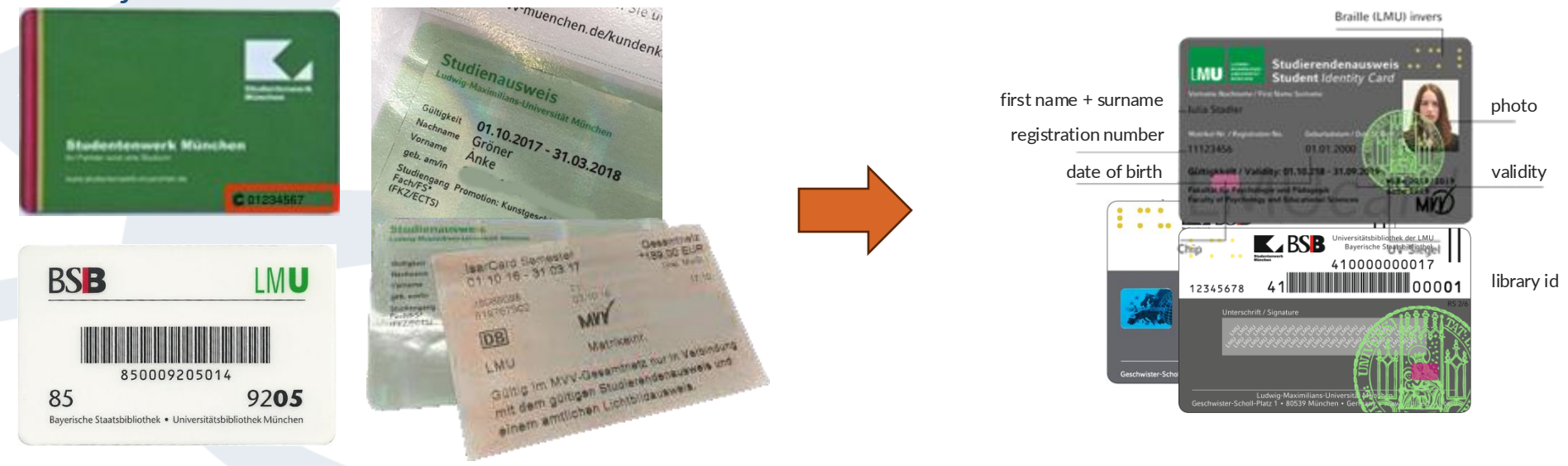


eduTAP

LMUcard – LMU Student ID

The transition of LMU Student IDs from paper to plastic cards (LEGIC CTC4096-MM410)

→ A Project 2015 – 2019



Goal: A modern integrated, contactless card for all services at LMU ← Better User Experience

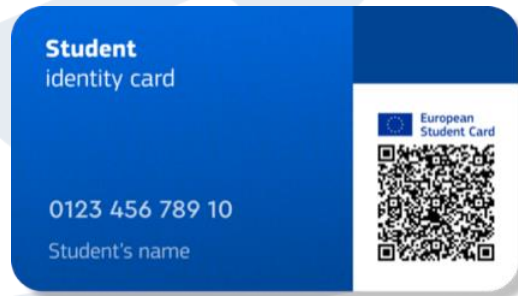
- ✓ Visual Student ID – contactless readable registration number
- ✓ Payment at mensa / canteens, cafeterias and vending-machines, copiers
- ✓ Library ID ✓ Public Transport Ticket (MVV Semesterticket)

It was a hard project with a lot of stakeholders

The European Student Card (ESC)

The **European Student Card** – a project to add a **European dimension** to local Student IDs

✓ Securely validate the student status



Idea by →
Jean-Paul Roumegas
(les Crouse)

European Student Card logo

Combines the European identity (text & flag) with the QR code. It facilitates the implementation and recognition of the ESCs

ESCN - European Student Card Number

Unique card identifier that serves to link a specific card to the specific higher education institution who produced it and to identify a card

ESI - European Student Identifier

Unique identifier that allows to link a given student to a specific ESC. It is not recommended to be printed on the card.



European Student Card



✓ A great vision

LMUcard + European Student Card (ESC)

LMU as an international university

International Partners and the European University Alliance

Strategic partnerships

- LMU-Bordeaux Research Cooperation Program
- Cambridge LMU Strategic Partnership
- LMU-China Academic Network (LMU-ChAN)
- LMU-NYU Research Cooperation Program
- LMU-TAU Research Cooperation Program
- LMU-Todai Cooperation in the Sciences
- LMU-UCB Research in the Humanities

EU projects – The European University Alliances

European University Alliance for Global Health (EUGLOH)



Global collaborations

- LMU - China Scholarship Council (CSC) Program
- LMU-Harvard Young Scientists' Forum
- LMU's Latin America Network



International alliances

- League of European Research Universities (LERU)
- European University Association (EUA)
- Europaeum



Infinity partners → infinity stakeholders → infinity systems



EUGLOH - The European University Alliance for Global Health

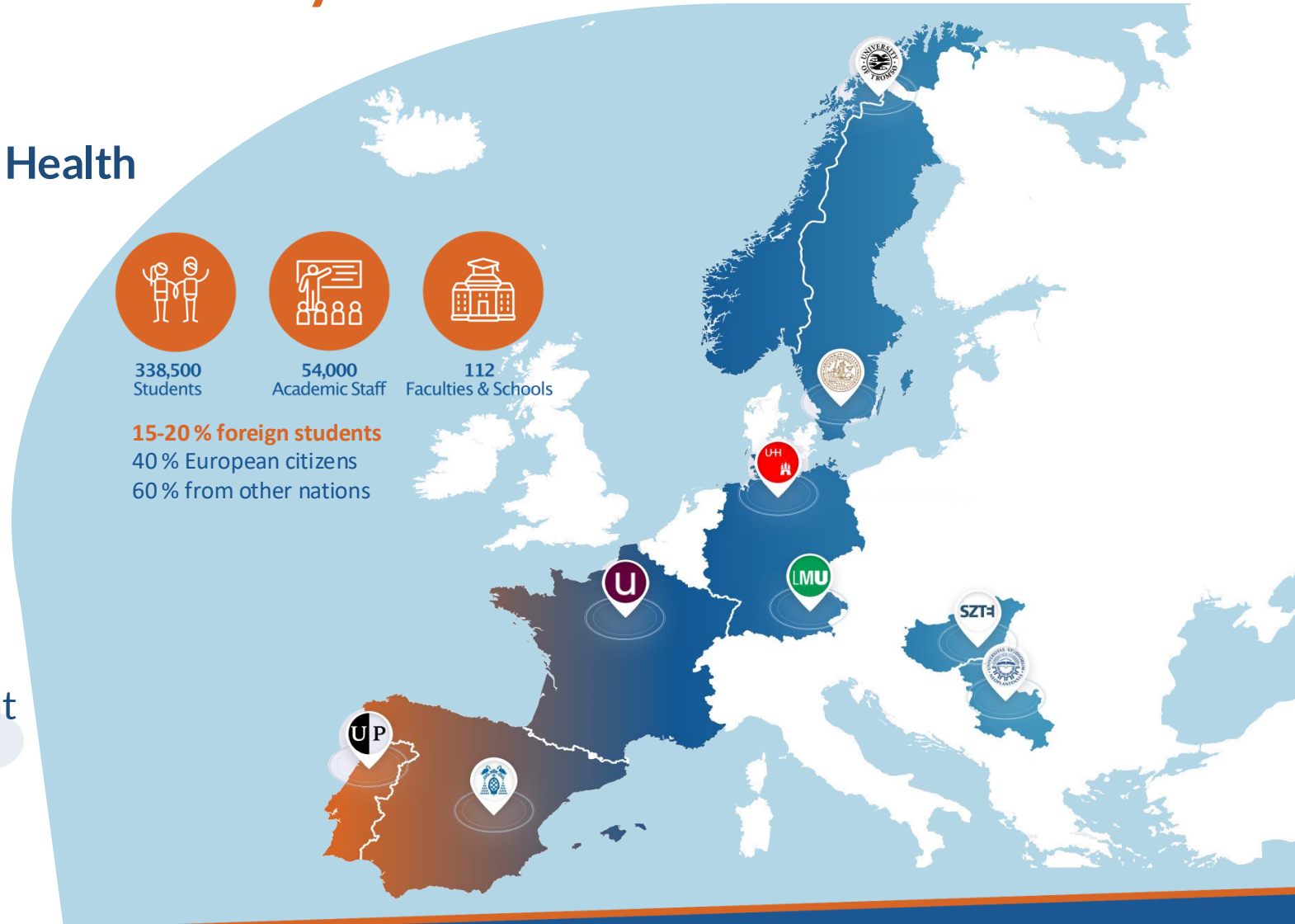
9 Universities, 1 Goal

Building Europe's Campus for Global Health

- Equipping students and staff with the skills needed to shape the future of global health
- Creating a **seamless, inclusive European campus** that enables **mobility**, joint learning opportunities, and **shared infrastructure**
- Addressing pressing health issues through interdisciplinary, co-created education, research and innovation that engages with social needs



15-20 % foreign students
40 % European citizens
60 % from other nations



A shared vision – seamless mobility

EUGLOH & the European Student Card have a **shared vision**

- ✓ Seamless mobility
- ✓ Easy service access

For:

- ✓ students,
- ✓ staff and
- ✓ professionals



EUGLOH ESC Working Group
formed end of 2020:

All EUGLOH universities should **adopt ESC**
and define services to share (Lead Paris-Saclay).

But differ in the audience and baseline
technical requirements

The European Student Card (2020)

Basic concepts

The European Student Card (ESC) is a new standard created by European Higher Education Institution (HEI) to promote student mobility within Europe (EHEA zone).

→ The long-term objective is to give all European Students the same rights as a local student when travelling abroad.

So, the idea is to create a card that can be accepted anywhere within Europe. This card gives access to an unlimited number of services that students use in their daily life, such as borrowing book from university library, accessing student cafeteria with discounted prices, accessing university premises, using public transportation etc.



Short- and long-term perspectives

The first basic service to be set up is **student status authentication**. This service offers the possibility for any organization (private business, public administration, association, HEI...) to confirm whether a card holder is a student or not. **This service is a necessity because student status authentication is usually a prerequisite to give access to other services that are restricted to students.**

- **Other services will become available afterward**, such as:
 - The possibility to use the card as an **electronic wallet** for day-to-day expenses
 - The possibility to write new information on the card after it has been delivered, granting access to a wide range of new services for the student
 - The possibility of using the card as a **key** to allow **exchange of data between HEIs**, to facilitate administrative procedures during student mobility
- **Dematerialisation of the card itself, using a smartphone**

Source: ESC Handbook for institution: Users's Guide to connect to the Platform
(<http://europeanstudentcard.eu/wp-content/uploads/2017/02/ESC-User-Guide-def.pdf>)

EUGLOH campus cards state and connected services

→ Vision:

All university members should be able to use services of host university while on visit → easy and contactless



Status of four core members (IT wise - 2020)

Ludwig-Maximilians-Universität München

- Legic CTC Cards (Legic Advant + Prime)
- Payment (Studierendenwerk München)
- Cloud-Printing and Scanning (Secure-Printing and Payment)
- Library
- Ticket for Public Transport System

Université Paris-Saclay

- NXP Mifare DESfire Cards
- Payment (izly – les Crous)
- Library
- Physical Access Control

Lunds universitet

- NXP Mifare Classic Cards
- Physical Access Control (online and offline)
- Login at copiers (for copy/print/scan)
- Library
- Several other services, but not any payment services

Universidade do Porto

- NXP Mifare Classic Cards / Java Cards with Mifare Classic emulation
- Library
- Physical Access Control
- Canteen & cafeterias
- Printing Service
- Attendance

All contactless, but
Not compatible technologies

EUGLOH & ESC – a shared vision and ambitions

We have a shared vision: **seamless mobility & easy service access**

But an ESC QR-Code was no option for us (GDPR & IT-Security wise), the long-term perspective a digital smartphone pass looked fascinating



→ a great vision



EUGLOH **wanted to share their services** across the alliance, and beyond (as we all have more partners)



The IT-Persons from LMU, U.Porto and Lund already knew each other → we met at ECCA 2019

- ✓ **understand the technological possibilities,**
- ✓ **knew and understand the topics, have access to major technology providers**
- ✓ **but also, the capabilities of the ESC Project team**

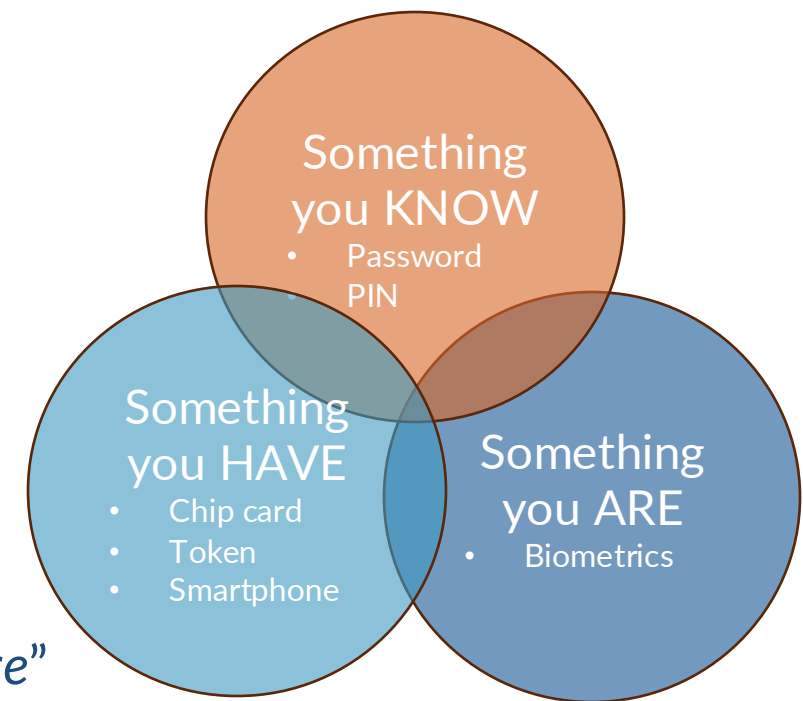
→ *If we want a good and interoperable solution in the near future, we must come up with a working proposal*

→ *That meets our requirements and audience (students, staff and professionals)*

Key to all Services → Identification / Authentication

- Triple-A-System
 - (Identification)
 - **Authentication**
 - **Authorization**
 - **Accounting**
- **On-site services access** is about proving your entitlements and identity
- “*something you know*” as an authentication method for on-site service access is not very **efficient**, best a combination of “*something we have*” + “*something we are*” → Smartphone Wallet passes + authentication via biometrics
- We already have a powerful IDM System for **online-services** → Shibboleth → eduGAIN-AAI

Authentication Factors



Identification / Authentication for Service Access

- ✓ **Interoperability happens on the reader side** → contactless communication protocols are better than visual codes
- ✓ Almost all readers in the market could **read more than one standard** (*maybe a firmware upgrade necessary*)
- ✓ QR-Codes are **not considered a secure** identification element (especially with an URL as content)

→ It is all about User Experience



Campus Cards are about Accessing Services

Identification

- **Identification**
- **Status verification**
- **Proof of entitlement**
- Single Sign On / 2FA / FIDO2
- Attendance check
 - Check-in for exams
 - attendance at courses
 - Time recording
- Electronic Signature (of legal documents)

Electronic payment / cashless campus

- Canteen & cafeterias
- Vending machines
- Printing / scanning (secure & follow me printing)
- Ticketing (events and conferences)

Physical Access Control

- Areas (campus, parking lot)
- Buildings
- Rooms
 - Classrooms
 - Labs
 - Computer rooms
 - Learning spaces
 - Offices
 - Accommodation facilities (i.e., dorms)
- Athletic facilities

Library services

- Access to / borrow
 - Physical media (book, audio and video media)
 - E-media (book, audio and video media)
- Learning spaces

Transport

- On campus services (university shuttle service)
- Public transport tickets & discounts

Discount and promotions

- **Discounts** on cultural activities
 - Museums
 - theaters
 - cinemas
- Shops
- Restaurants

On- & Off-Campus → On-Site Usage / Proximity Use-Cases
+ Online Services



Campus Cards from a User Experience Point of View

Campus Cards (Student IDs / Staff IDs / Affiliate IDs) function as:

- **Official (visual) Identity Documents** and for university branding
- Primary medium for **Service Access** on-site → **proximity use-cases**
- Service Domains:
 - On-Campus
 - **Identification** and attendance tracking (courses, internships, exams)
 - **Payment** (open loop or closed loop payment) and discounts eligibility
 - **Physical Access Control** (buildings, labs, dorms)
 - **Library Services**
 - on-campus & public **Transport** integration (regional semester ticket models)
 - Off-Campus (Museums, theaters, shops, restaurants, ...)
 - **Status-based Identification** for third-party services
 - **Eligibility Verification** (e.g., student status, age-based discounts)



Campus Cards are about Accessing Services

The Service Domains

- Payment / Cashless Campus,
- Physical Access Control ,
- Library services,
- Transport

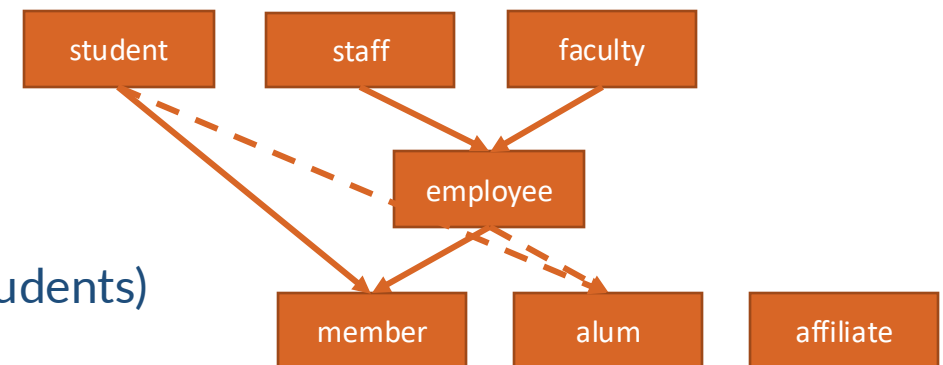
are “**local**” services, which only needs to be solved on a local level

- ✓ Separate service passes can choose best fitting technology
- ✓ Services are provided to **ALL** persons (not only one role, e.g., students)

→ Student IDs / Staff IDs are **legally limited** to enrolled / employed members of a Higher Education Institution (HEI)

→ Dedicated service passes can be legally issued to not enrolled / employed persons ← Users of the service

Educational role system (from eduPerson)



Trust level / Assurance Level

- Very important for **IDENTIFICATION & AUTHENTICATION**
- Quality of identity (personal) data
- Type of authentication & consequences for identification medium

no trust

normal

Authentication:
 • username / email + password
 Identity proofing:
 → self-asserted
 → verified e-mail address

substantial / medium

Authentication
 • certificate based
 • FIDO2
 • Smart ID
 Identity proofing:
 → officially confirmed
 → manual check of a government issued photo-ID

high

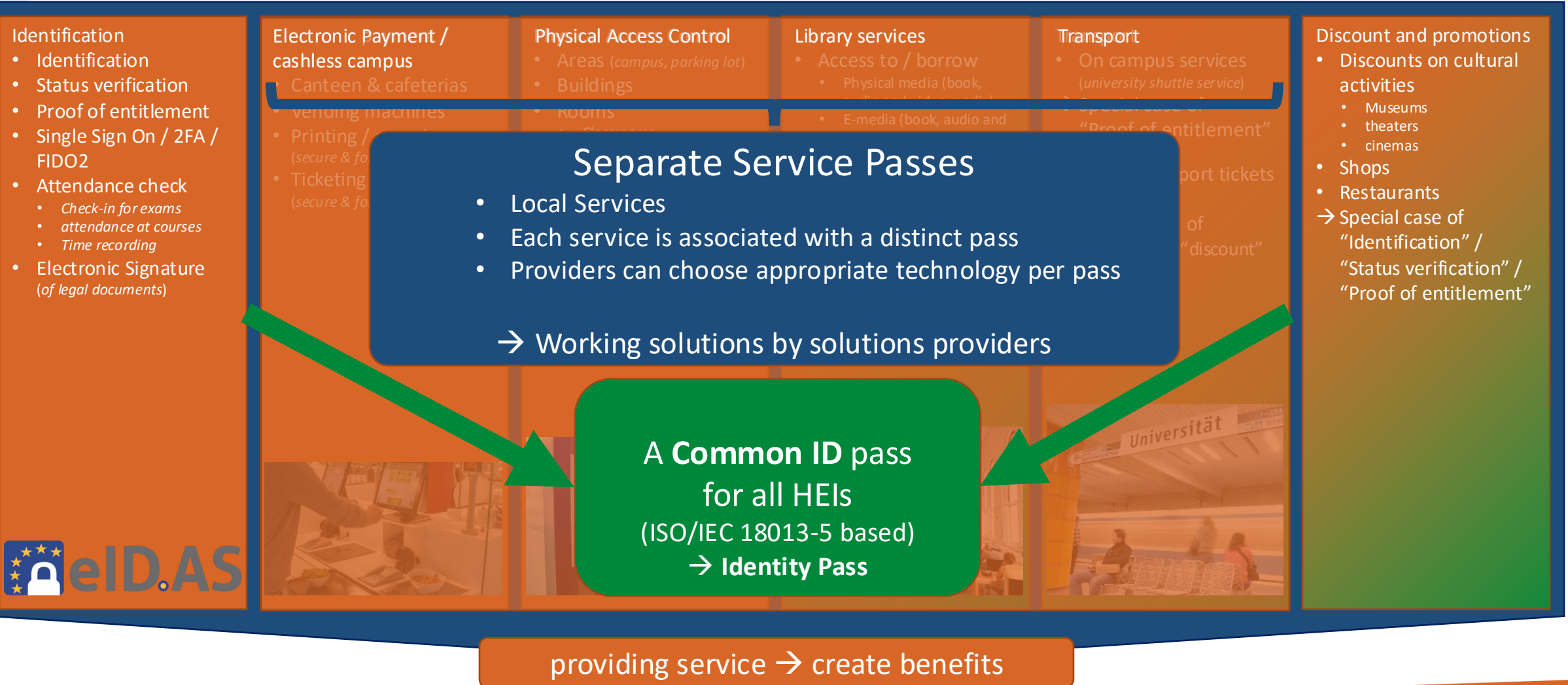
Authentication
 • **eID.AS** compliant
 → eID of EU Citizen IDs
 → provided by HEI for foreign Students
 Identity proofing:
 → officially confirmed
 → Identity document automated and technically checked to be genuine and represent the claimed identity

Account Creation vs. Account Authentication

- Do not need to be on same trust level / assurance level
- **Constraint:** Trust Level of Account Creation $\geq$ Account Authentication
 - Authentication requiring a higher trust level than for the creation does not make sense
- Example: LMU Library Card
 - Trust Level for *Issuing Library Card* is **Substantial/Medium** due to legal requirements (Allgemeine Benützungsordnung der Bayerischen Staatlichen Bibliotheken*)
 - ID card
 - Registration address
 - Trust Level for *Book Lending* is **normal** (via a credential – NFC or Barcode)

* <https://www.gesetze-bayern.de/Content/Document/BayABOB>

Think outside the card: Medium → Services



Campus Cards are about Accessing Services

Almost all services are **SHARED** services

- ✓ University library ← Can be used by members of all universities in town / metropol area, in a university alliance or are open to the public
- ✓ Transport (semester ticket) ← Are just discounted and time limited public transport tickets (valid for 6 month)
- ✓ Canteens, cafés and vending machines on campus ← Can be used by all roles and members of all universities
- ✓ Physical Access Control (PAC) ← For all roles in a university / university alliance

Services are provided by **different Service Providers** (European universities are not business focused)

- ✓ Canteens, cafes and vending machines are operated by Student Affairs Unions
- ✓ Transport services are operated by local transport companies
- ✓ University Buildings (PAC) are university or state owned

→ Each **Service Provider** can issue their **own pass**, matching their needs and security requirements

Munich campus as an Example of SHARED Services



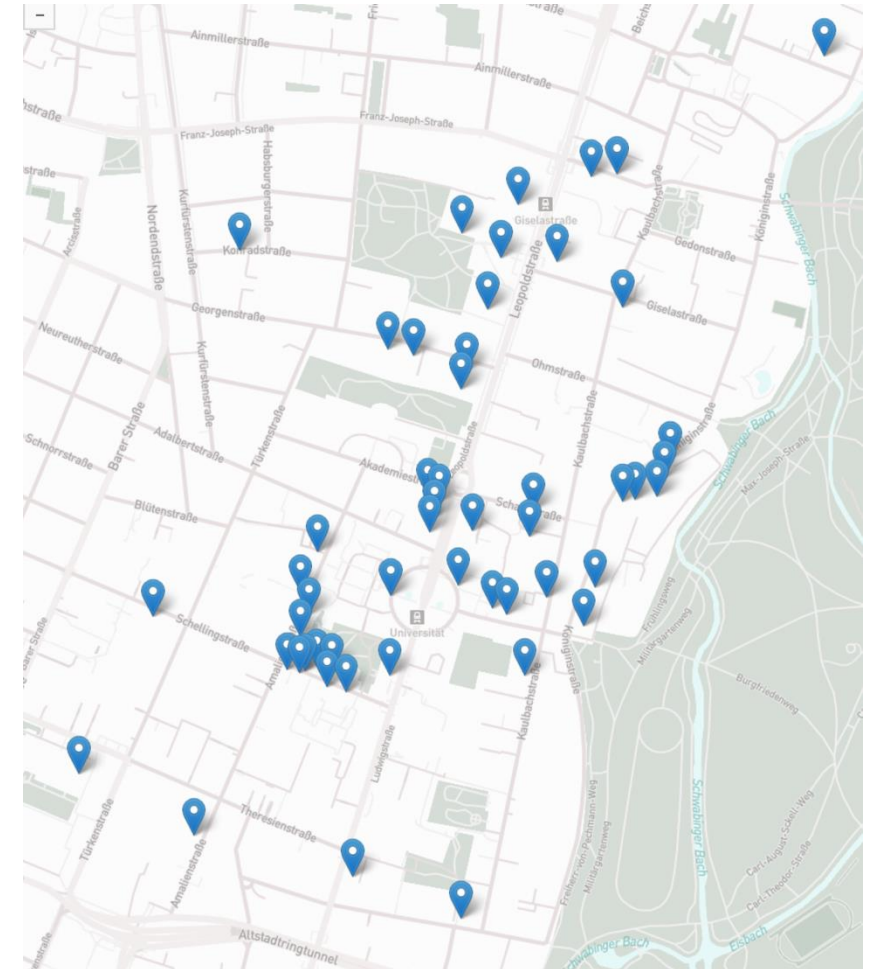
Most European Universities do not have a clearly demarcated campus area. They are just in the city.

The university is not even the main service provider for student life.

Typical Service-Providers:

-  (=Student Affair Union)
 - Canteen & cafeterias
 - Vending machines
 - Student housing / dorms
 → Responsible for all universities in Munich (LMU, TUM, HM, ...)
- Libraries are dedicated (almost independent) units at LMU we have two libraries: University Library and
 

 Bavarian State Library
- Transport → Public Transport System





European University Alliance for Global Health

eduTAP

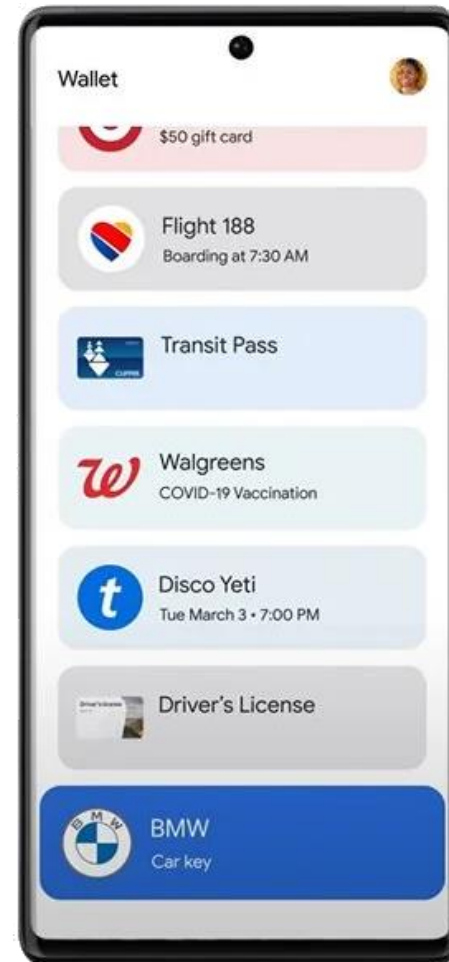
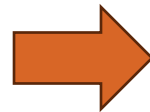
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Concept of eduTAP



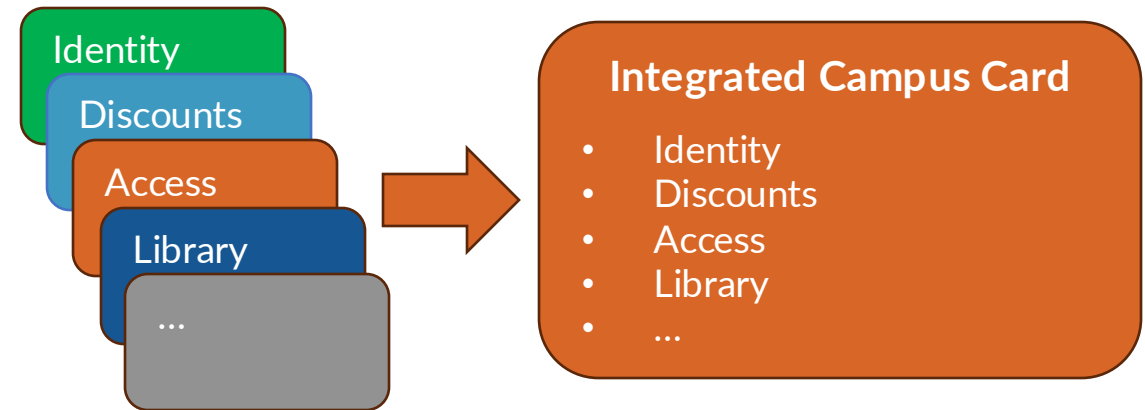
Apple / Google Wallet – Focus on User Experience



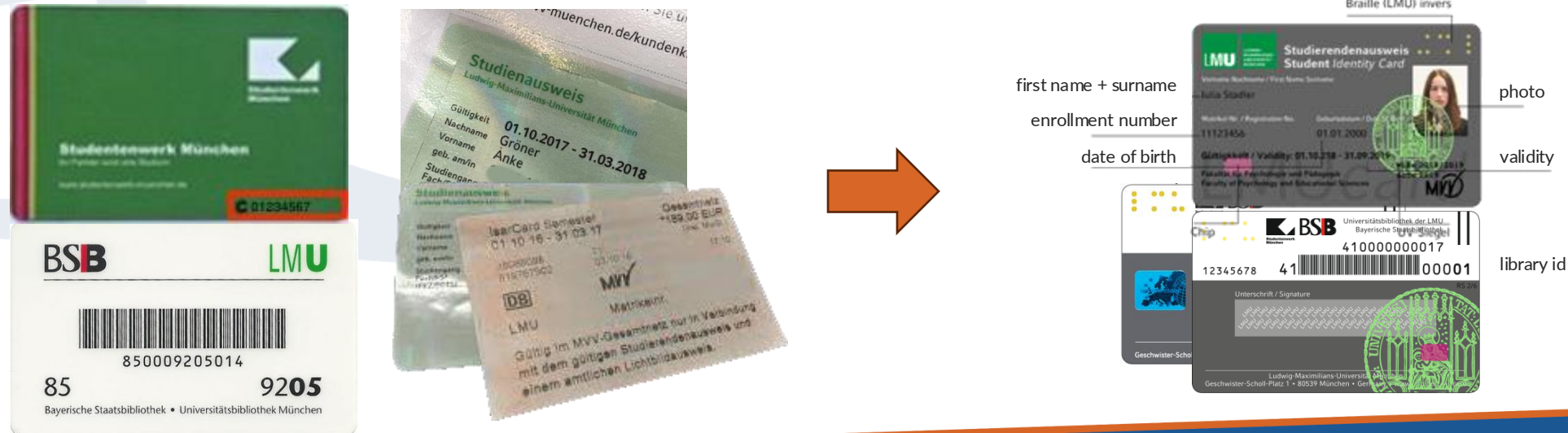
Campus Cards from a User Experience Point of View

User preferences for **Physical Wallets**:

- ✓ Less cards are better
- ✓ **Contactless** cards are better



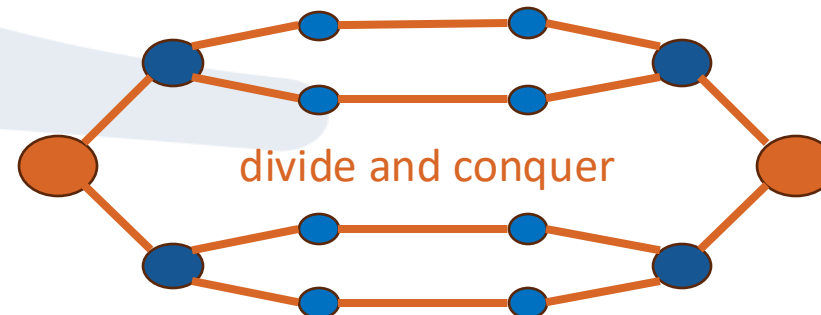
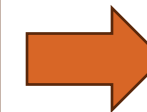
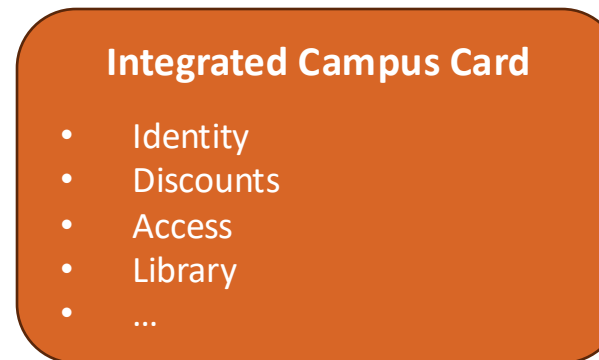
Example LMU 2015 – Transition from multiple cards into **ONE** integrated Campus Card



Campus Cards from a User Experience Point of View

User preferences for Smartphone Wallets:

- ✓ NFC / Contactless passes are better than scanning visual codes
- ✓ **One pass per service** is better
 - User control over which services to use
 - Auto-Presentation of pass needed
 - Clear contextual communication (additional information on pass specific for the service, specific branding of service)



Digitizing cards **vs.** Leveraging the possibilities of smart devices

→ Stages of Digitalization – Period of Digital transformation

Digitalization 1.0

- Digitization – create digital versions of analog documents

→ Digitization vs. Digitalization (machine readable content / payload)

Digitalization 2.0

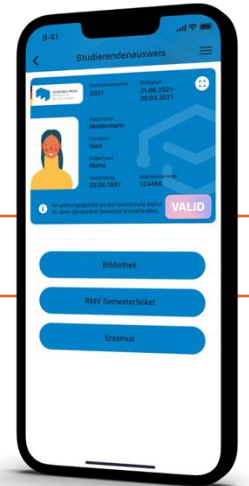
- Digitalization with internal focus
- Optimize internal administrative processes – **organization centric optimizations**
- Processing of data

→ Digital transformation (a hard shift)

Digitalization 3.0

- Digitalization with focus on the user – **user centric optimizations**
- Optimization on how services are discovered and seamlessly used
- Flow of data

→ Service-oriented
→ **Digital identity**
→ Interoperability



→ User Experience

eduTAP – educational tapping

tap to ...

- ✓ identify yourself,
- ✓ verify your status,
- ✓ proof your entitlement,
- ✓ claim a discount,

tap to access a service on-site directly:

- ✓ tap to pay (open and close loop-payments),
- ✓ tap to open a door,
- ✓ tap to lend a book,
- ✓ tap to take a campus shuttle.

TAP TO PAY



no app, just tap!

<https://eduTAP.eu>

eduTAP – educational tapping – What is eduTAP?

- eduTAP the concept
 - Splitting integrated campus cards into **dedicated service passes** and a **harmonized identity pass**
 - Leveraging the full potential of digital wallets
 - **Use existing**, secure and proven contactless **standards** – do not invent new standards

→ Focus on **User Experience** and **interoperability**

- eduTAP the software – to enable HEIs and Service Providers to:
 - create, issue, and manage passes within the smart device wallet
 - interact with wallet passes to give access to services, identify, and verify the status of students and staff members

→ eduTAP helps to fulfill the
great vision of the European Student Card
and **enables interoperability** and a **better user experience**

- ✓ It is all about **providing services** → **eduTAP is for anyone**
 - ✓ For users of the (home or host) institution
 - ✓ User incoming for a short-term or long-term mobility
- ✓ Staff / faculty members are more likely on short-term mobility than students
 - **leading by example**



UNESCO ISCED:2011:
International Standard Classification of Education

eduTAP – educational tapping – for the users

eduTAP passes enable using smart devices to perform transactions with a

simple tap

making everyday interactions **easier** and more secure both on campus and beyond.

→ User Experience

Our mission is to provide an **interoperable solution** that works for everyone

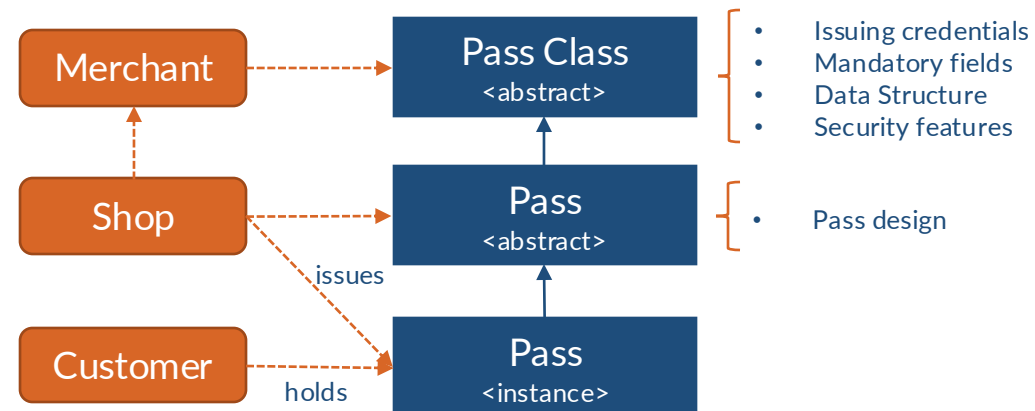
- for users of the (home or visiting) institution
- for user incoming for a short-term or long-term mobility
- staff / faculty members are more likely to be on short-term mobility than students

Technical details and features of Wallets / Passes

The wallet is a **container for multiple passes**

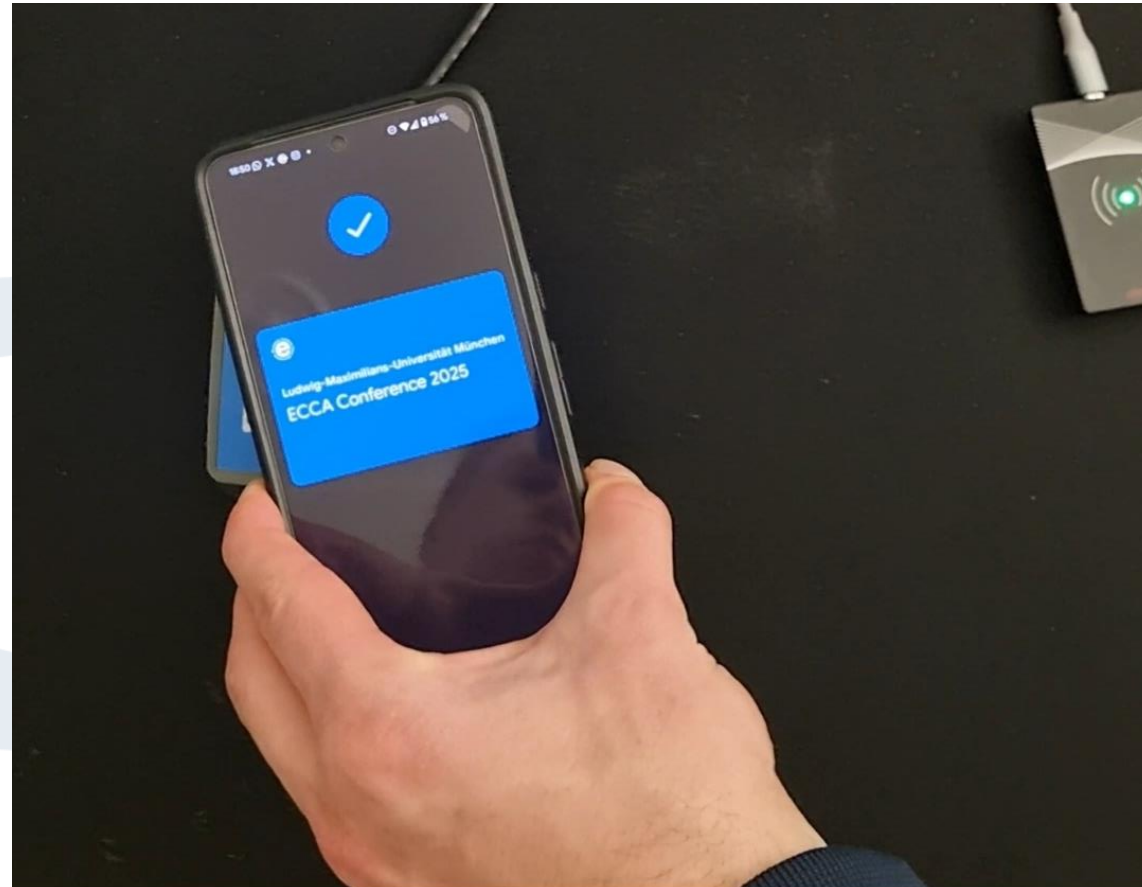
Technical features:

- **Auto-Presentment / Smart Select** ← reader selects the specific pass to interact with (if available)
- Selectable Authentication requirements ← **security features**
- Express-mode (Apple)
- **Battery reserve** (pass is still usable if phone has powered off; important for keys, transit tickets, payment)
- Groupable
- Validity dates
- Remote updateable / revokable
- **Offline usable** (→ non traceable)
- *Host Card Emulation possible*



Technical details and features of Wallets / Passes

Auto-Presentment / Smart Select of an NFC enabled Pass in the Wallet



Technical details and features of Wallets / Passes

Battery Reserve Mode – Use the Wallet even if it's powered of

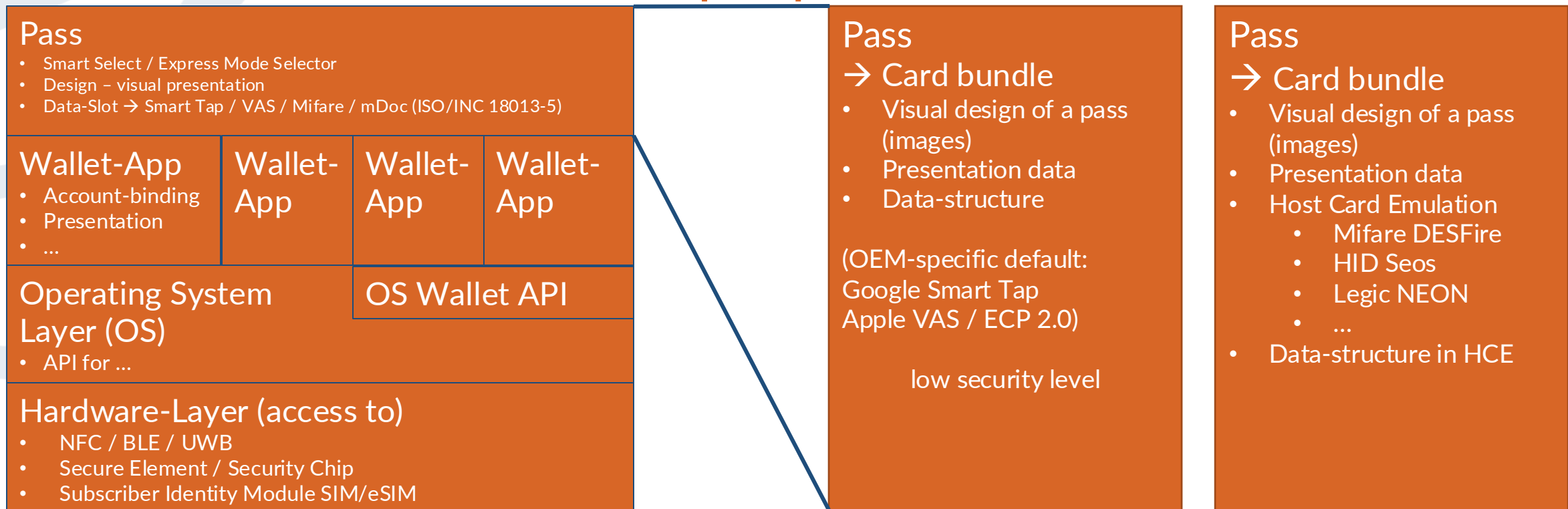
- for Keys (Dorm-Rooms, Class-Rooms, Building Access, Cars)
- for Payment
- for Transport Tickets



Up to 7 Days after shutdown of the smartphone

Technical details and features of Wallets / Passes

- ✓ Technical details and features of wallets / passes → Path to an **interoperable** campus card
- ✓ If each pass (technology) can be individually selected and leveraged, and the pass in the wallet can be selected by the reader (**Smart Select / Auto-Presentment**)
→ **One pass per service**



Technical details and features of Wallets / Passes

- For contact less communication is NFC the protocol
- Communication via ISO 14443
- Data Formats (File System access)
 - Google Smart Tap
 - Apple VAS (Value Added Services)
 - NFC ECP 2.0 (Enhanced Contactless Polling)
 - Host Card Emulation (HCE: of proprietary Protocols)
 - Mifare DESfire (via Mifare2GO / NXP secure element)
 - Legic NEON / Advant (via Legic Connect)
 - HID Seos
 - Internal the Secure Module acts as a JCOP smart card
- ISO 18013-5 Personal Identification
 - NFC handover to BLE or WiFi
 - Core functionality in Android and iOS OS API*



Security Level
Chip / Smartcard

Security Level
Smartphone

high

high

Legic NEON
Legic Advant
Mifare DESfire
HID Seos

ISO 18013-5

ECP 2.0

Smart Tap
VAS

low

Mifare Classic /
Card UID

low

- Security consideration required by use case

Technical details and features of Wallets / Passes

Apple and Google have good ideas, but

One size
does **NOT**
fit all.

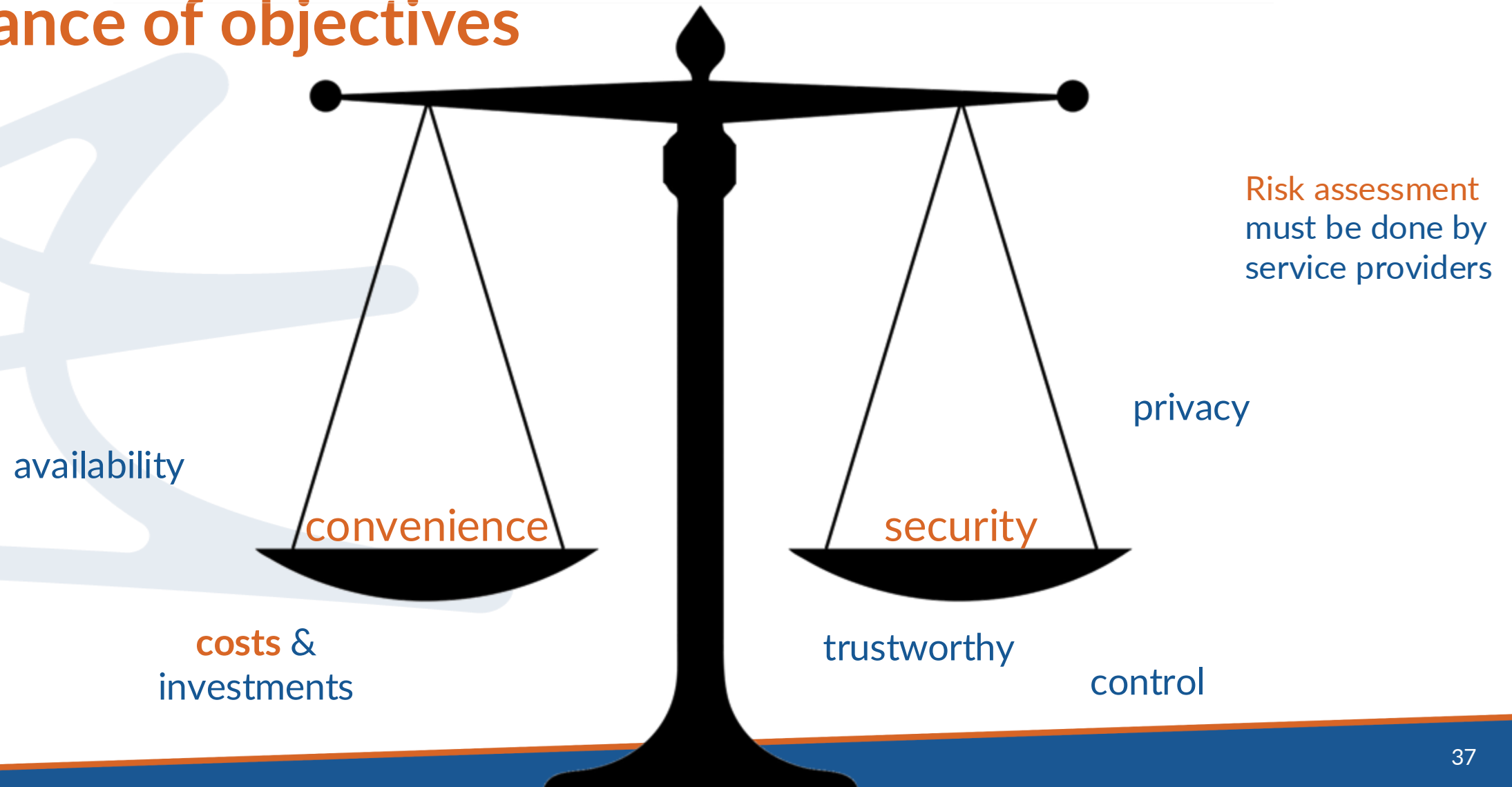


ONE SIZE DOES
NOT FIT ALL



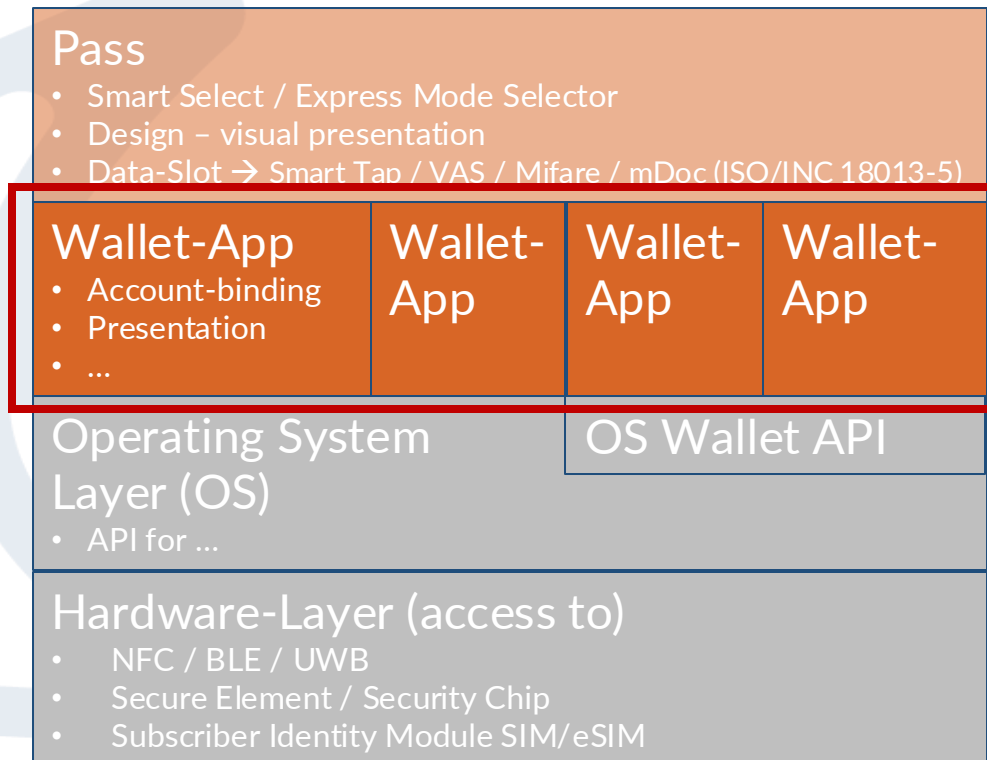
Technical details and features of Wallets / Passes

Balance of objectives



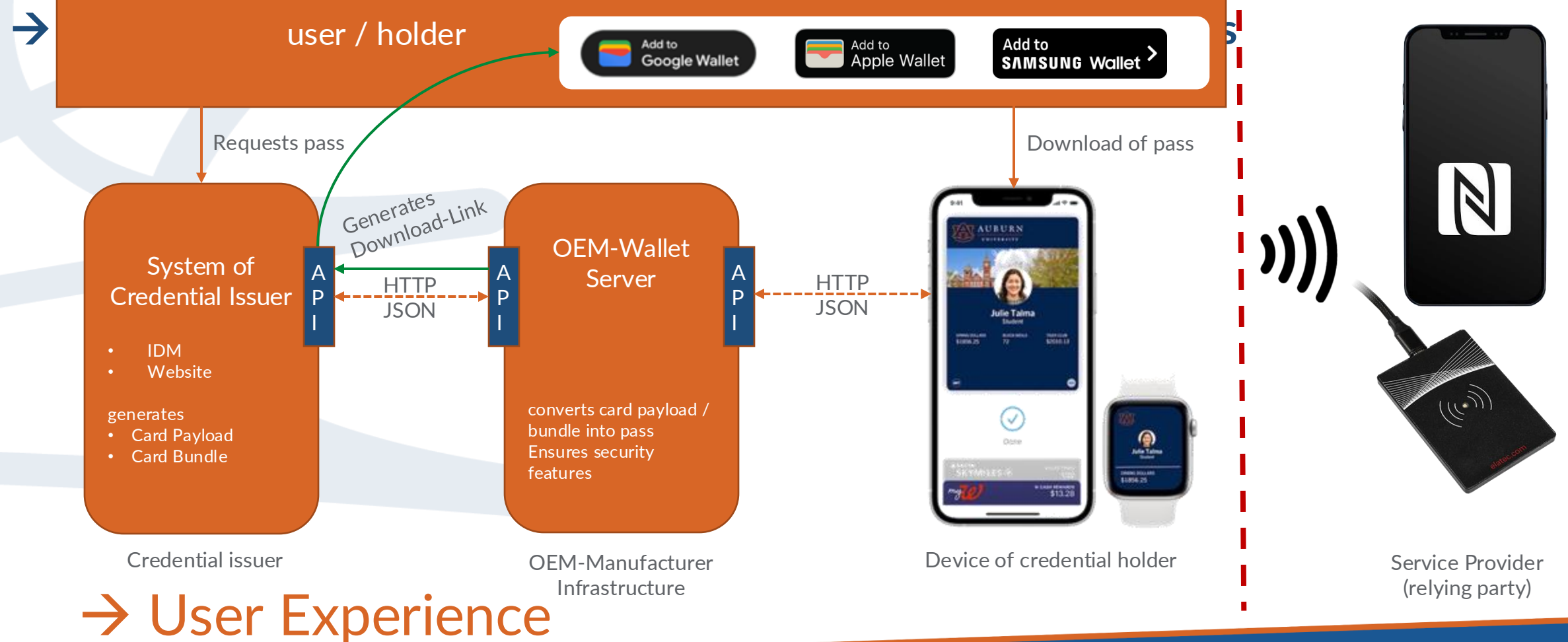
Technical details and features of Wallets / Passes

Detail view on the Wallet



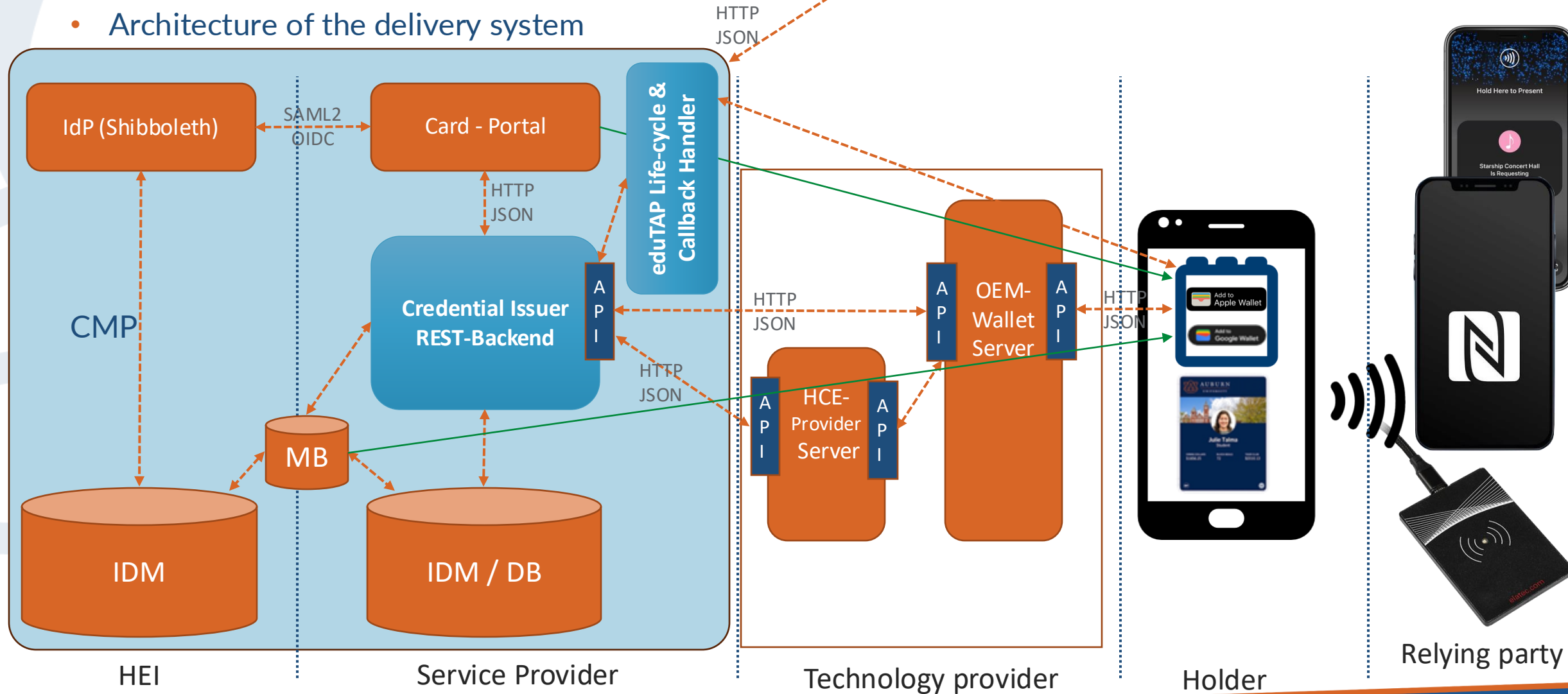
- The eduTAP solution is wallet **agnostic**, not bound to a specific wallet provider or company
- eduTAP just enables HEI and Service Providers to easily **create** and **issue** wallet passes

Issuing process of wallet passes



Credential Management Platform and eduTAP Components

- Architecture of the delivery system



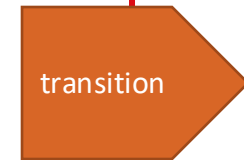
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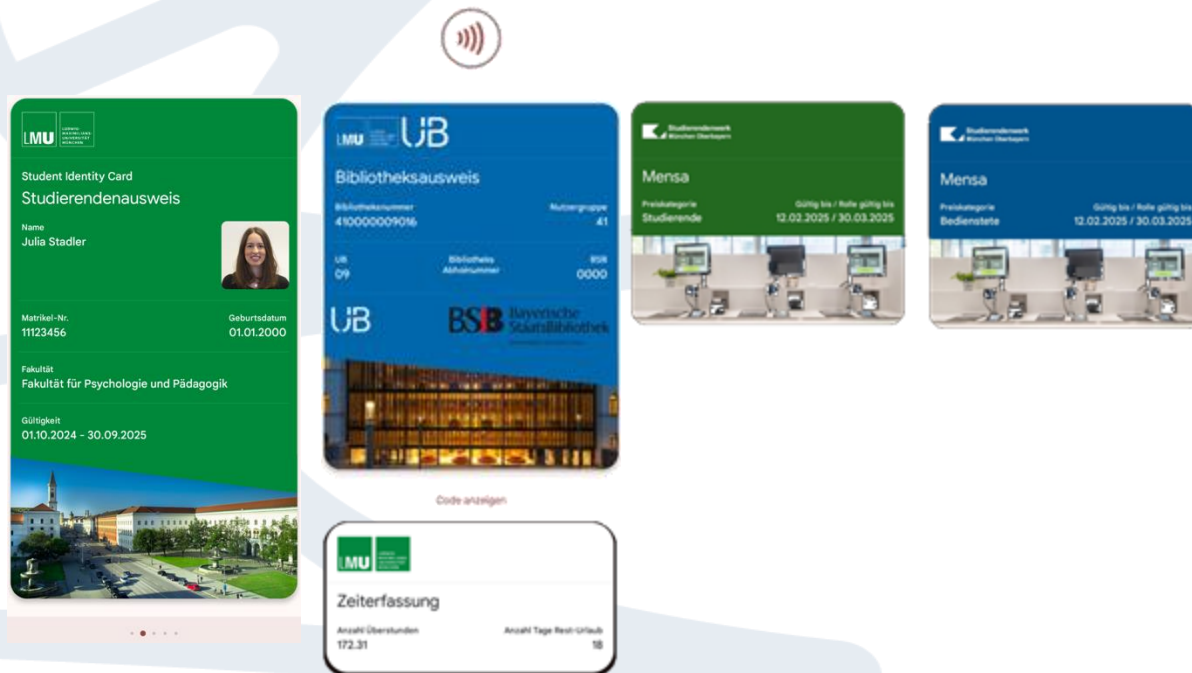
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Transition of credentials into the Smartphone Wallets

Examples: Active and potential IDs for LMU Munich

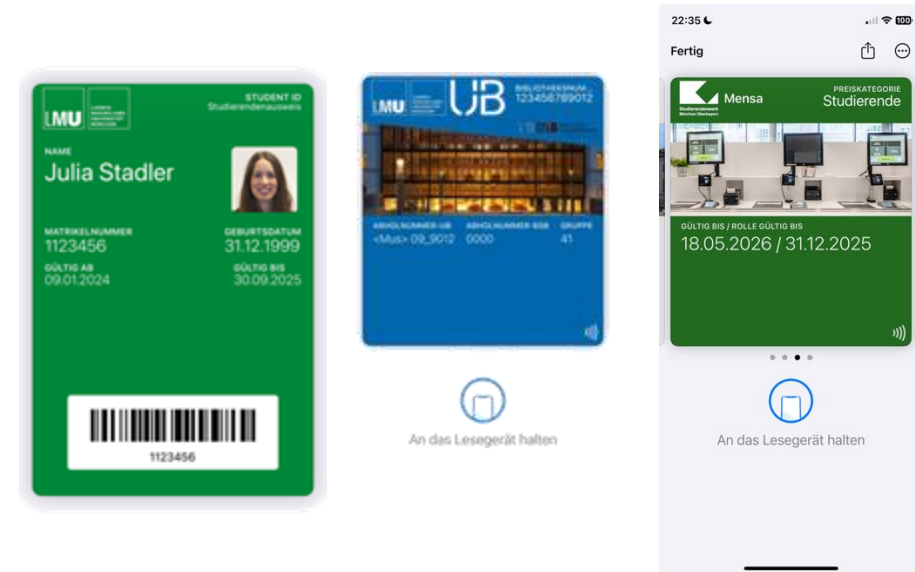
Google Wallet



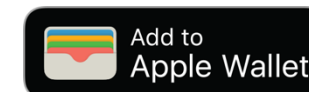
→ all **NFC enabled**



Apple Wallet

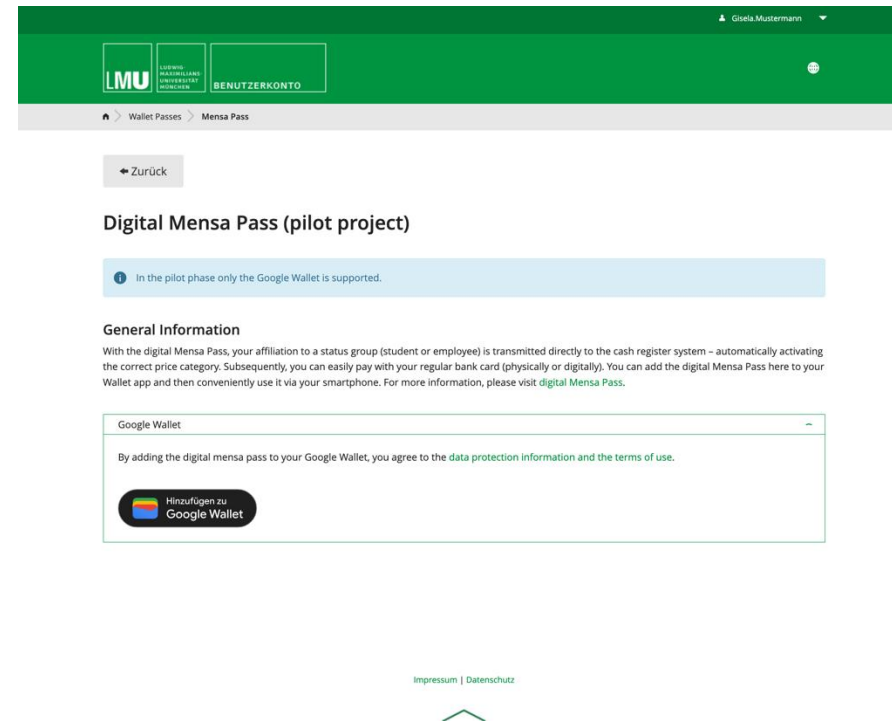
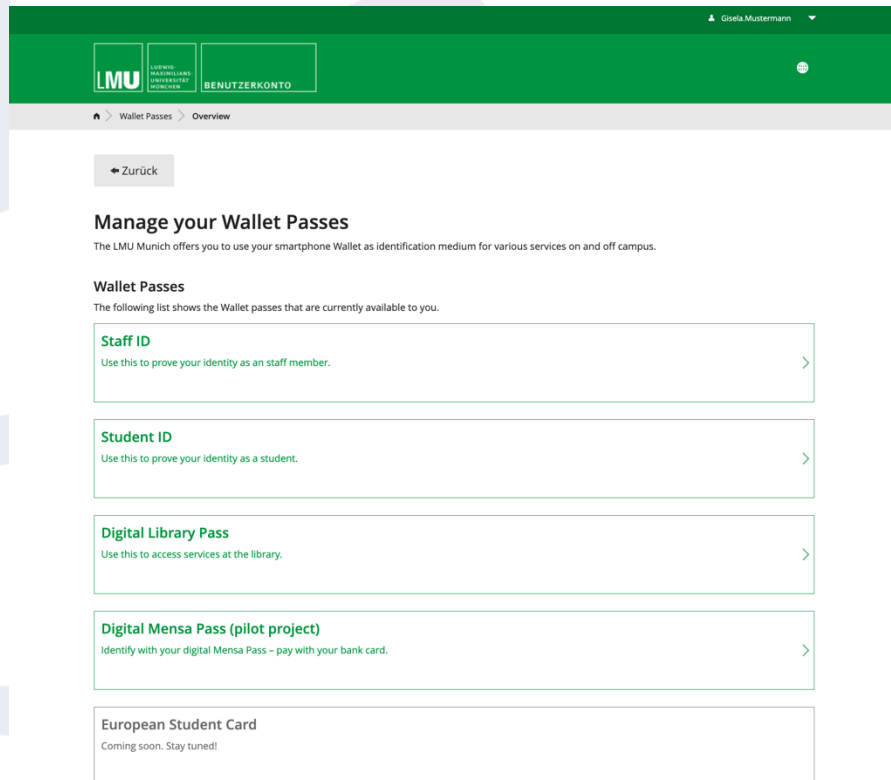


→ **NFC enabled** where possible

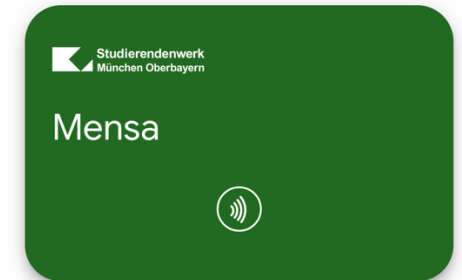


Transition of credentials into the Smartphone Wallets

Examples: LMU Munich Benutzerkonto – Pass Portal



Add loyalty card



After adding your loyalty card to Wallet, you'll see your point balance and loyalty benefits in places like Maps, Shopping, and more. You can turn this off in [Google Wallet passes data](#) or on an individual pass in the Google Wallet app.

Pass providers can automatically add related passes, promotions, offers and more to your existing passes which can be controlled on your device in Wallet settings.

[Terms and Privacy Policy](#)

Add

→ Plan to have all pass types online at LMU in the summer semester 2025 mostly design questions and information on the back are discussed

Transition of credentials into the Smartphone Wallets

LMU Student ID

The new **LMU Student ID** in the Wallet

- ✓ will come operational in summer semester 2025 (optional to LMUcard)
- ✓ for summer semester 2027 either physical LMUcard or wallet ID



Google



Apple

more wallets to come

Transition of credentials into the Smartphone Wallets

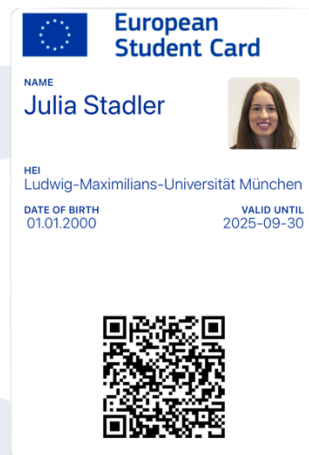
The European Student Card

The European Student Card

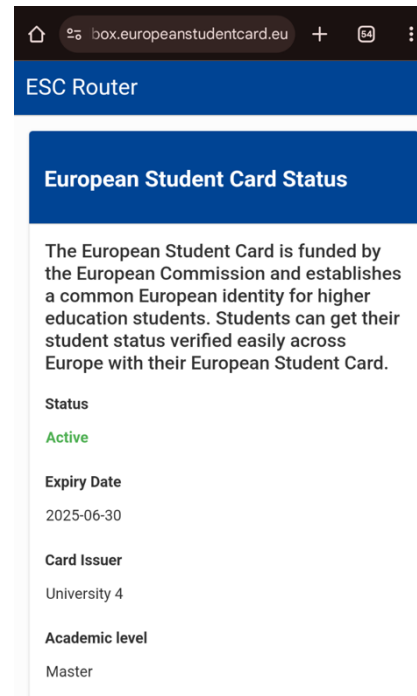
- ✓ Discounts off-campus
- ✓ European dimension → Erasmus+ marketing



Google



Apple



Transition of credentials into the Smartphone Wallets

Library ID

NFC enabled by default



Code anzeigen

Google



An das Lesegerät halten

Apple

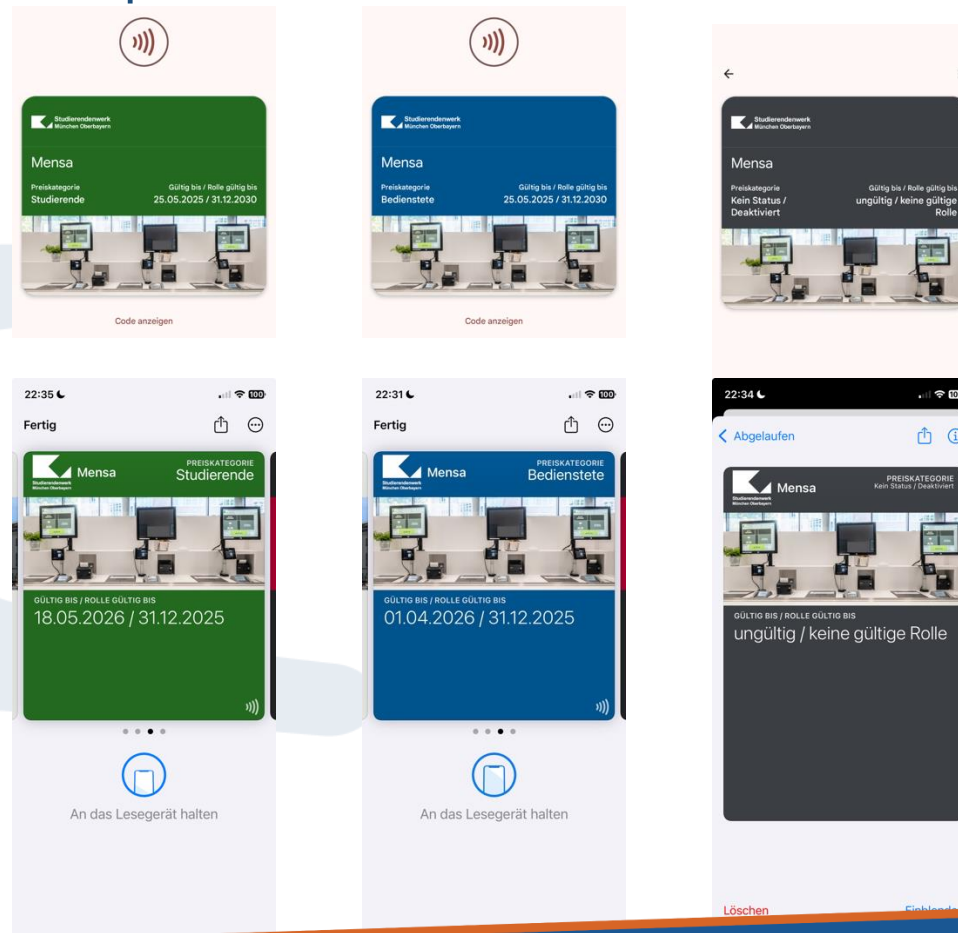


Discount-Pass for Mensa Studierendenwerk München Oberbayern



Eligibility verification for discounts with the Apple / Google Wallet via NFC, payment with an Open-Loop Bank / Credit-Card

Google Wallet
Since Feb. 2025



Apple Wallet

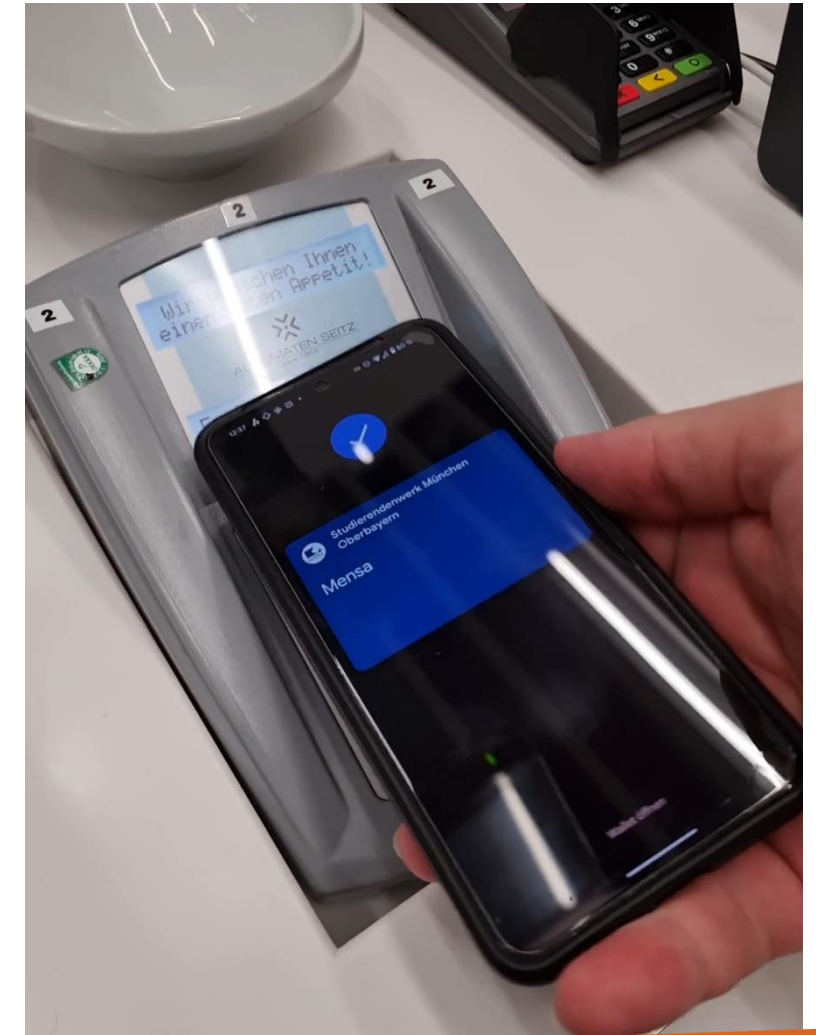
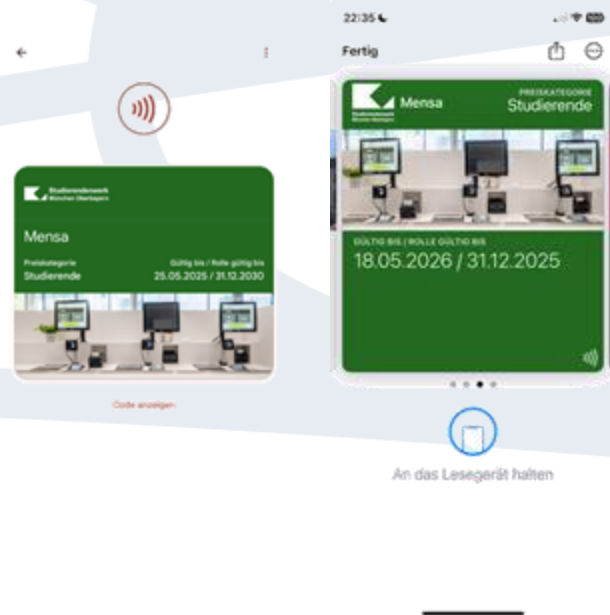
Discount-Pass for Mensa Studierendenwerk München Oberbayern



Payment Process in the Mensa

Double Tap:

1. Eligibility verification of status group (Student, Employee, Guest)
→ Discount will be applied
2. Pay with an Open-Loop Bank / Credit-Card



Transition of credentials into the Smartphone Wallets

Use Case: Physical Access Control (Access to learning spaces)

Physical access to the learning spaces (provided by the library)

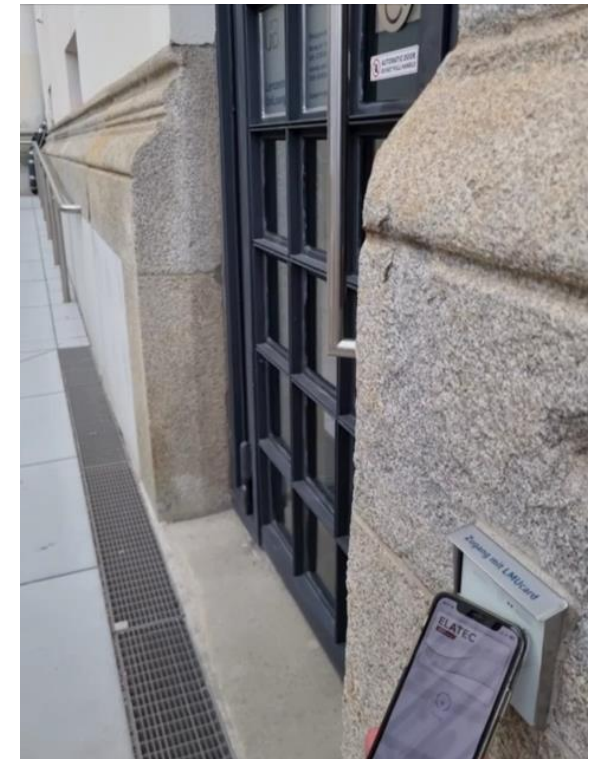


LMUcard

Access



Google Smart Tap



Apple

TEVALLID



European University Alliance for Global Health

eduTAP

A *sneak peek* into the future of Student IDs

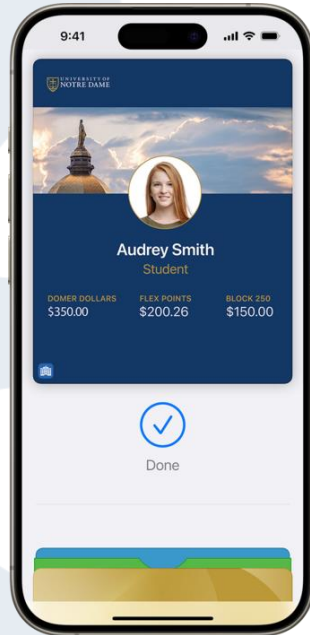
→ More interoperability and less passes by Verifiable Credentials

eduTAP Common ID



eduTAP and the current Apple / Google Student IDs

How about the native, current **Student IDs** in Apple Wallet & Google Wallet (Mifare or HID based)?



- ✓ Fantastic integrated cards (**one card** for all services, perfect for **corporate**)
- ✓ *Should be used for all university services* (Apple's intention)
 - University / organisation-centric optimization
 - Does not support the “seamless mobility” requirement
 - Shared services, how about other users than own students?
 - **LESS secure** (if following the ESC data zone approach)
 - Not the required User Experience, and does not meet the legal requirements

This is a **dead-end solution**, it will not help or bring benefit, it actually provides a **WORSE User Experience**

It does NOT support **Selective Disclosure** and **Zero Knowledge Proof**

Apple and Google is aware of this and working on it!

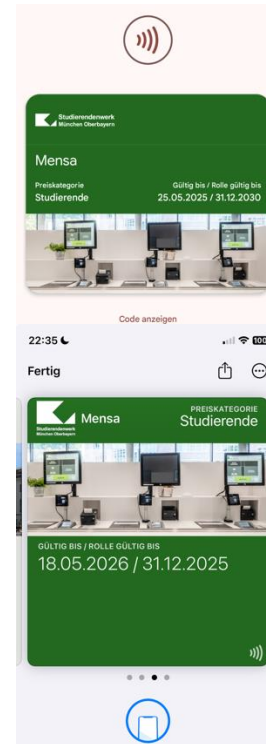
Apple and Google Access Platform are good for PAC and Payment passes

- **NOT for Student or Staff IDs**
- **GOOD dedicated service passes**

Wallet Passes are about a great User Experience

A **great User Experience** is key to adoption and usage

- ✓ Non-NFC Passes are **NO GOOD** User Experience
- ✓ Unnecessary passes should be avoided ← NO dedicated pass for simple eligibility verification (status, institution affiliation)



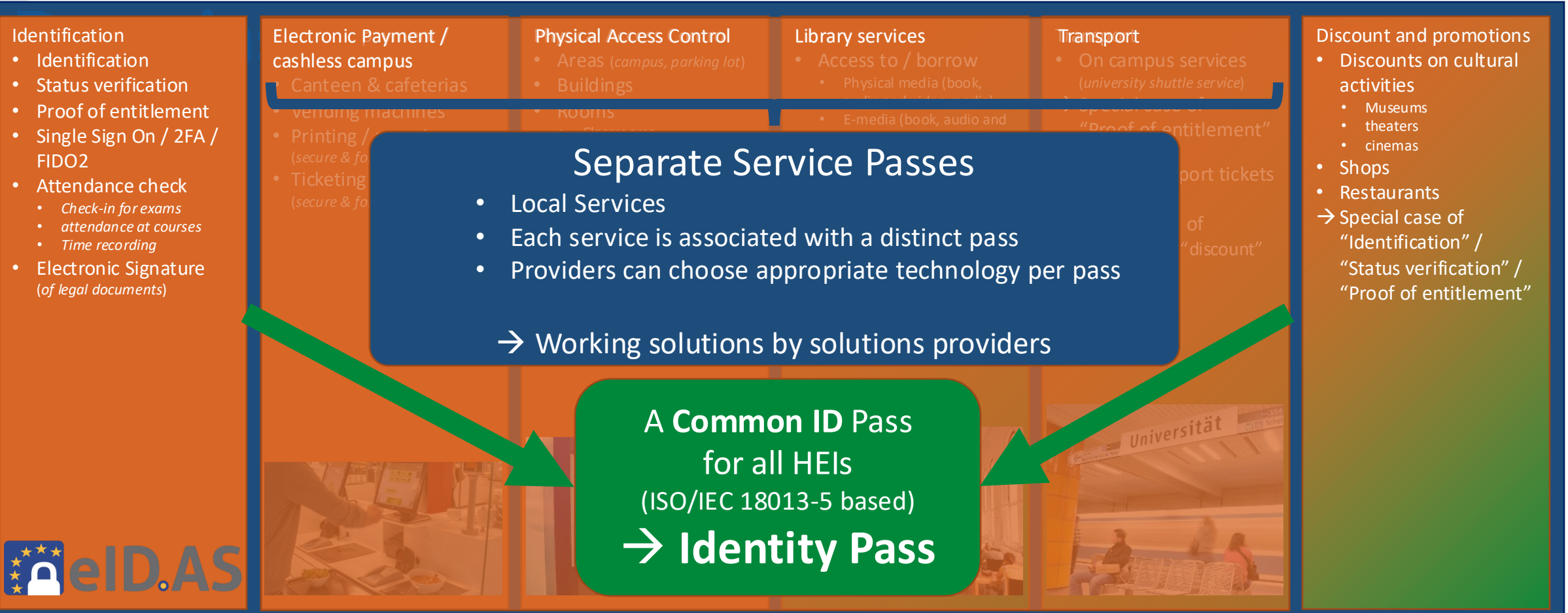
transition

Is there a better solution?



- ✓ Verifiable Credentials
- ✓ Selective Disclosure
- ✓ Zero Knowledge Proofs

Think outside the card: Medium → Services



Service Domains: Identity and Discount

From a legal point of view, Identity Documents of public universities are **official state photo IDs**

- ✓ Legal requirements
 - ✓ Security considerations
 - ✓ High assurance level
- { → **International standards** for HEI Identities → eduGAIN (incl. eduPerson, SCHAC)
 e.g. European Student Identifier → routable information for course participation at other institutions (during mobility programmes)

Identity Documents (Student ID, Staff ID and Affiliate IDs) are used

On-Site (proximity use-cases)

Online

On-Campus

Off-Campus

- ✓ Attendance at courses
- ✓ Check-in for exams
- ✓ Time recording
- ✓ Discounts in canteen & cafeterias
- ✓ Identification for university authorities (Student legal office and examination offices, campus security)

- ✓ Identification for public authorities (e.g., police)
- ✓ Identification for third party (e.g., checkin at hostels)
- ✓ Eligibility verification for discounts and access (e.g., role-based, age-based, institution related)

- ✓ As authentication factor

A question of scale: Student IDs

European Union

- ~ 5,000 Higher Education Institutions (HEIs)
- ~ 18,800,000+ tertiary students (2022)

Erasmus+ Countries

- Includes EU + Iceland, Liechtenstein, Norway, North Macedonia, Serbia, Turkey
- Significantly expands the European higher education landscape → ~ 20,000,000+ tertiary students

Worldwide

- ~ 18,000 recognized HEIs across 180+ countries
- ~ 200,000,000+ tertiary students globally

→ Organized in
National Research and Education Networks (NRENs)



→ It should work everywhere: seamless mobility & easy service access
→ a GLOBAL Student ID ←



GÉANT – The IT provider for Higher Education in Europe

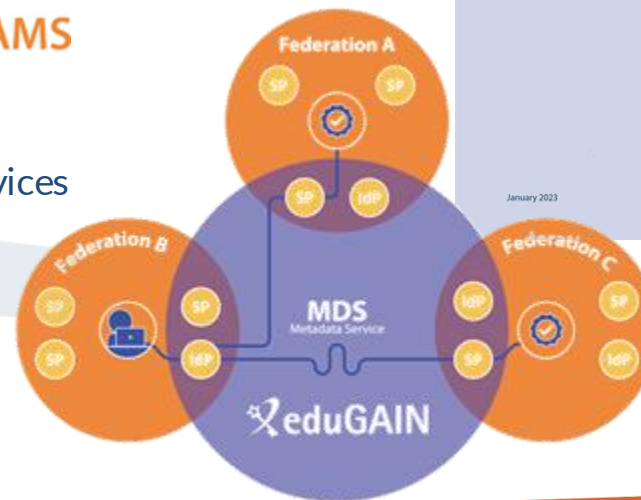
The GÉANT Association is the collaboration of European National Research and Education Networks (NRENs)

- Non-Profit Organization
- Service-Provider for IT / e-Infrastructure services for the European Higher Education Sector
- Pan European – Non EU members
- Key Services:

- **GÉANT Trust & Identity Services**



- GÉANT Network and Connectivity Services
- GÉANT Security Services
- GÉANT Cloud Services
- Procurement Agency for EU-HEIs



→ eduTAP is a perfect fit for GÉANT Trust & Identity Services

Identity Pass: Reuse Shibboleth / eduGAIN data schema: The common authentication method in the educational sector

- ✓ Shibboleth – local or in federation (national or multi-national) → eduGAIN-AAI → SAML2 or OIDC



**LMU**

LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

DEUTSCHES FORSCHUNGSNETZWERK
AUTHENTIZIERUNGS- UND AUTORISIERUNG-INFRASTRUKTUR

LOGIN





English version

Ludwig-Maximilians-Universität München

— Anmelden bei LMU Zoom —

Benutzername

Password

☐ Anmeldung nicht speichern

☐ Übermittelte Daten anzeigen

Anmeldung

Bitte verwenden Sie zur Anmeldung

- Ihren **LMU-Benutzernamen** (= Campus-Kennung, mit oder ohne Domain **@lmu.de**, **@campus.lmu.de** oder **@med.lmu.de**) oder
- Ihre im **LMU-Portal** ersichtliche **LRZ-Kennung**.

Diese Single-Sign-On-Lösung **Shibboleth** erlaubt Ihnen den einheitlichen Zugang zu diversen Webanwendungen der LMU und anderer Universitäten innerhalb des **Deutschen Forschungsnetzes (DFN)**.

Bei der erstmaligen Anmeldung, oder wenn Sie die Checkbox **Übermittelte Daten anzeigen** auswählen, erhalten Sie eine Übersicht Ihrer personenbezogenen Daten, bevor diese an die Webanwendung übertragen werden.

Nach dem Login haben Sie so lange ohne erneute Anmeldung Zugang zu allen Shibboleth-fähigen Webanwendungen, bis Sie den Browser schließen. Zum Logout müssen Sie daher den Browser komplett beenden.

[Impressum](#) | [Datenschutzerklärung](#) | [Kontakt](#) und [FAQ](#)



LUDWIG-
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LMU

DEUTSCHES FORSCHUNGSNETZWERK
AUTHENTIFIZIERUNGS- UND AUTORISIERUNGS-INFRASTRUKTUR

LOGIN



Informationsweitergabe an MyAcademicID IAM Service

Beschreibung des Dienstes: *The MyAcademicID IAM Service provides Identity and Federated access management for the services of the European Student Card Initiative and the services directly supporting the digitisation of Erasmus+.* The MyAcademicID IAM Service is provided by GEANT using the eduTEAMS service.
Zusätzliche Informationen über diesen Dienst:
<https://wiki.geant.org/display/SN/MyAcademicID+Identity+and+Access+Management+Service>

An den Dienst zu übermittelnde Informationen

Anzeigename	
Zugehörigkeit	student member staff employee
Vertrauensgrad	https://refeds.org/assurance https://refeds.org/assurance/IAP/low https://refeds.org/assurance/ATP/ePA-1m https://refeds.org/assurance/ID/eppn-unique-no-reassign https://refeds.org/assurance/ID/unique https://refeds.org/assurance/ATP/ePA-1d https://refeds.org/assurance/IAP/medium https://refeds.org/assurance/profile/cappuccino
Berechtigung	urn:maced:dir:entitlement:common-lib-terms
Persönliche ID	@lmu.de
Zugehörigkeit	staff@lmu.de member@lmu.de student@lmu.de employee@lmu.de
eduPersonTargetedID	HV6nox/0SxHyzkH4z+vIGNdD80=
eduPersonUniqueid	7b719738ecae7b120b6383786b47fe98@lmu.de
Vorname	Simon
E-Mail	@campus.lmu.de
o	uni-muenchen
schacHomeOrganization	lmu.de
schacPersonalUniqueCode	urn:schac:personalUniqueCode:int:esi:lmu.de:
Nachname	

Datenschutzinformationen dieses Dienstes

Die oben aufgeführten Informationen werden an den Dienst weitergegeben, falls Sie fortfahren.

Wählen Sie die Dauer, für die Ihre Entscheidung zur Informationsweitergabe gültig sein soll:

☐ Bei nächster Anmeldung erneut fragen.

——Ich stimme zu, dass meine Informationen dieses Mal an den Dienst weitergegeben werden.

☒ Erneut fragen, wenn sich die Informationen ändern, welche an den Dienst weitergegeben werden.

——Ich bin einverstanden, dass dieselben Informationen in Zukunft automatisch an diesen Dienst weitergegeben werden.

Diese Einstellung kann jederzeit mit der Checkbox auf der Anmeldeseite widerrufen werden.

Ablehnen

Akzeptieren

Impressum | Datenschutzerklärung | Kontakt und FAQ

[illegible]

Digital Identity in the Wallet

Common Verifiable Credential Formats / Protocols for Digital Identities

- SD JWT (EU)
- ISO/IEC 18013-5 (mDL)
- W3C Verifiable Credential Data Model

Issuing and Redemption Protocols

- OID4VC
- OID4VP
- ISO/IEC 18013-5

For **proximity** use-cases NFC / BLE transmission is crucial

→ Our choice ISO/IEC 18013-5 (mDL) for Wallet credentials
JWT

Image Source: <https://www.linkedin.com/feed/update/urn:li:activity:7333379048112361472/>

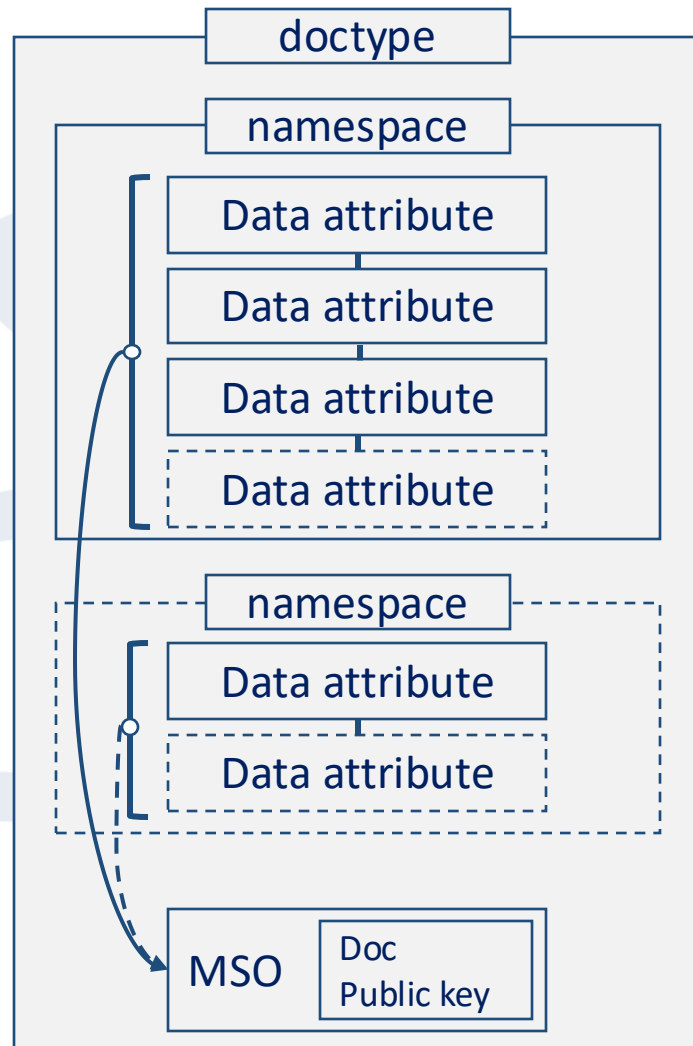


A Common Identity Pass using the ISO/IEC 18013-5 model

- The “ISO/IEC 18013-5 Personal Identification” model is designed to support
 - A protocol for two devices to establish a **secure wireless communication** channel and exchange structured request and response message
 - **Identification** of the credential holder (**user binding**)
 - **Selective release of data elements** by the credential holder (**data minimization / consent**)
 - **Pre-consent** mechanism for frequently repeated processes (e.g., discount payment at Student Affairs Unions)
 - A protocol to retrieve credential data directly from the mobile device of the holder, **purely offline**, facilitating **availability** and **non-traceability**
 - An **optional** protocol to **retrieve additional data** from the issuing authority
 - A mechanism to establish **integrity** and **authenticity** of the credential data
 - A mechanism to confirm **device binding** (signing at transaction time)
 - Multi-document and **multi namespace aware** ← extendable for other use cases
- **Established governmental credential format that is already in use** ← **Following the ARF of EUDI-Wallet**
→ mobile driving licence / eu.europa.ec.eudiw.pid.1 (upcoming national eIDs in Europa)
- **Already supported** by the OEM-Smartphone-Manufactures (Apple and Google)
→ see: <https://android-developers.googleblog.com/2020/11/privacy-preserving-features-in-mobile.html>



Identity Pass: ISO/IEC 18013-5 Data model



Reuse of directory data (common eduGAIN attributes) 

- portrait (photo, *mandatory for identification documents*)
- givenNames
- sn (surname)
- email
- eduPersonAffiliation / eduPersonScopedAffiliation
- eduPersonEntitlement
- eduPersonAssurance
- schacHomeOrganization
- schacPersonalUniqueCode → European Student Identifier (ESI) - MyAcademicID
- schacDateOfBirth
- schacYearOfBirth
- schacPlaceOfBirth
- schacExpiryDate
- ...



*works like a Shibboleth authentication
on-site → trained processes*

see also DC4EU education ID

(<https://code.europa.eu/ebsi/json-schema/-/tree/main/schemas/vcdm1.1/education/verifiable-education-id>)

Identity Pass: Path to the best User Experience and Interoperability

Current Status Apple:

Identity Documents are required (photo ID, sensitive personal data) → Options in Apple Wallet

- ✓ Apple VAS Pass (Generic) with visual codes (e.g. QR-Code) **no NFC**
- ✓ Apple VAS Pass Passes (Generic) with NFC (currently not allowed)
- ✓ Student ID (Mifare DESFire) ← for inter-university usage **unencrypted** (not secure)
- ✓ Identity Passes (currently not allowed)

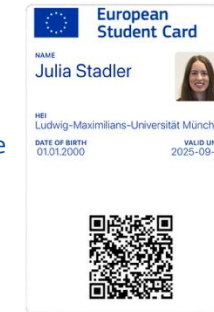
Not optimal

As possible data formats:

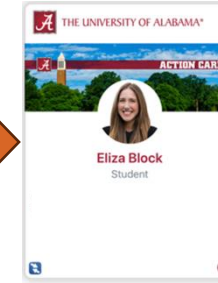
- Plain text / JSON
 - Encrypted JWT or similar
 - Verifiable presentation not possible
- All not optimal for GDPR

Optimal Solution: a Verifiable Credential / ISO 18013-5

- ✓ Works offline and untraceable
- ✓ All relevant data for identification and status verification
- ✓ Interoperable for identification and simple service access for inter-university usage
- ✓ A potential global solution
- ✓ Already implemented in Apple and Google Wallet



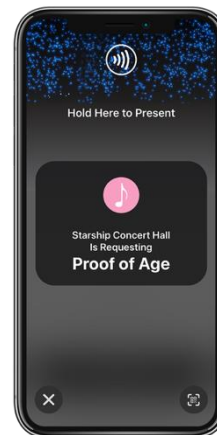
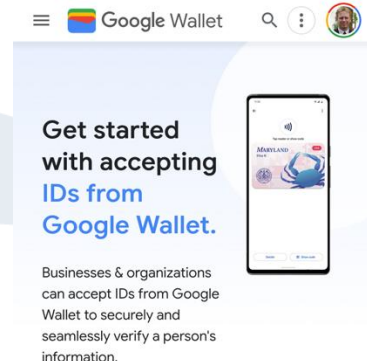
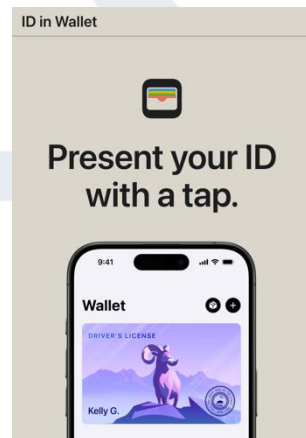
Phase 1
QR-Code
ESC



Phase 2
Apple Access Student ID with only one Mifare AID unencrypted → encrypted, signed JWT as payload eduTAP simple ID



Long-term goal
Verifiable Credential based on ISO 18013-5



→ Defined in ARF for European Digital Identity Wallet (EUDI-Wallet)



→ Identity passes would be the best solution for Student IDs, Staff IDs and Affiliate IDs

Identity Pass: Path to the best User Experience and Interoperability

Current Status Google:

Identity Documents are required (photo ID, sensitive personal data) → Options in Google Wallet

- ✓ Google Smart Tap Passes (GenericObject) with visual codes(e.g. QR-Code) or NFC
 - ✓ Google Smart Tap Passes (GenericPrivateObject) with NFC (no Image)
 - ✓ Student ID (Mifare DESFire) ← for inter-university usage **unencrypted** (not secure)
 - ✓ **Identity Passes** (currently not allowed)
- Not optimal**
- As possible data formats:
- Plain text / JSON
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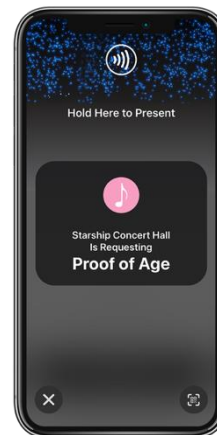
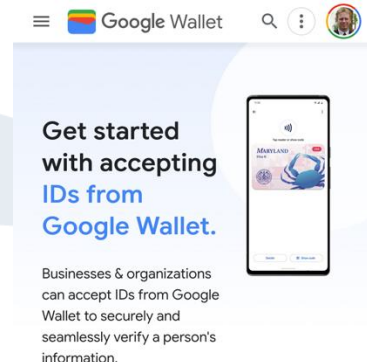
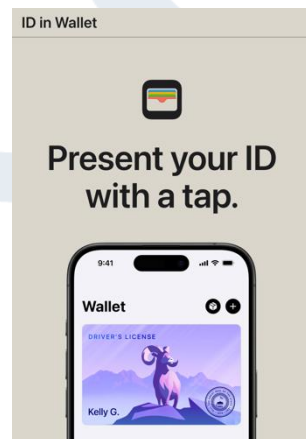
Phase 1
QR-Code
ESC



Phase 2
NFC JWT
eduTAP simple ID



Long-term goal
Verifiable Credential
based on ISO 18013-5

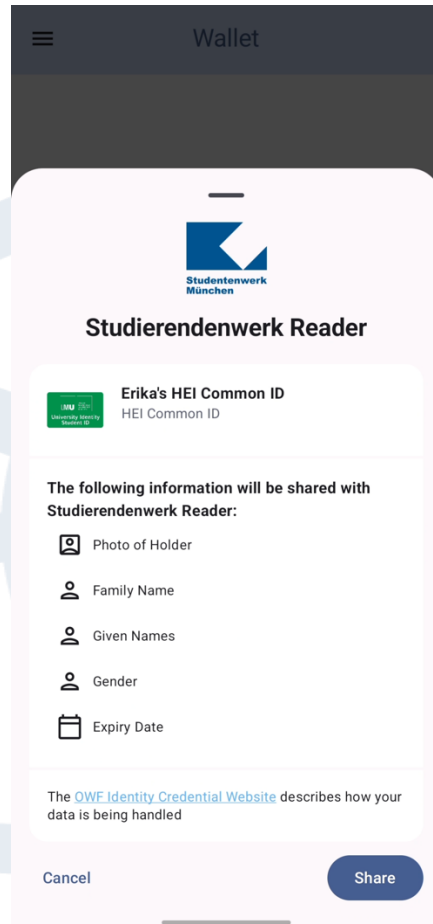


→ Defined in ARF for European Digital Identity Wallet (EUDI-Wallet)

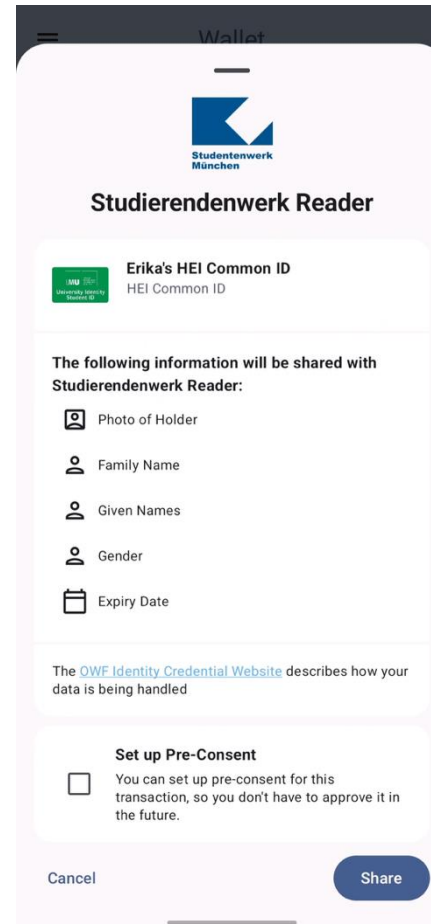


→ Identity passes would be the best solution for Student IDs, Staff IDs and Affiliate IDs

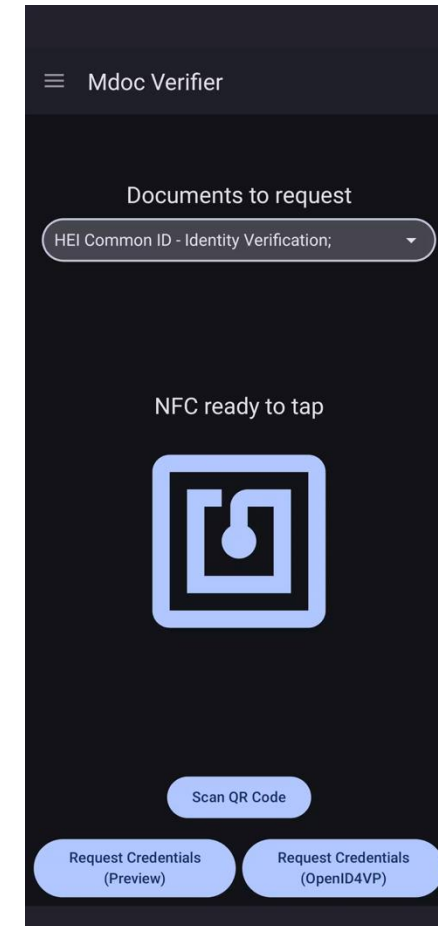
Identity Pass: Path to the best User Experience and Interoperability



Informed Consent



Informed Consent
+ Pre-Consent



Verifier App

Campus Cards – Point of Views

Technology Experts vs. Consumers (Students, staff and professionals)

The Technology and issuing processes are not important to the consumers (card / pass holders)

The IDs and service passes should:

- ✓ **just work**
- ✓ be immediately available when needed
- ✓ be secure & GDPR compliant
- ✓ trustworthy
- ✓ **accepted everywhere**
- ✓ *be boring – no need for attention*

Students, teachers and researches should be able to focus on the important things,

→ their studies and research

→ enjoy student mobility (and do not need to wait weeks for a physical campus card)

“Boring” is a good thing

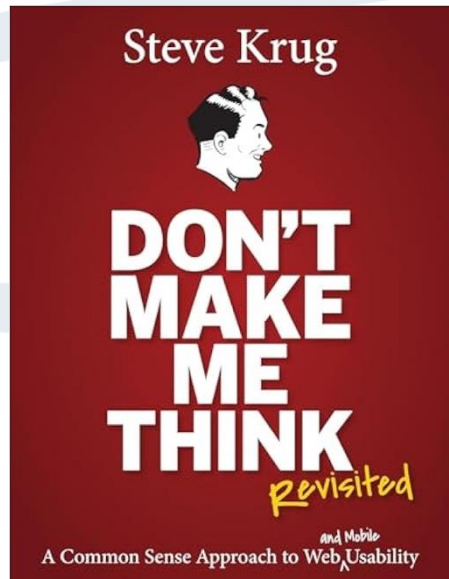
Every company gets about three innovation tokens.

Source: Dan McKinley, "Choose Boring Technology"

<http://mcfunley.com/choose-boring-technology>

“Boring” let you get things done

Same is true for people, if you **need to think** to get something done it costs effort, and energy for other more important things



Don't make me Think

→ User Experience is important

You should **NOT** need to care about “boring” and “un-important” stuff

You should be able to **focus on the important** things
(studying, learning, researching, ...)





European University Alliance for Global Health

eduTAP

<https://eduTAP.eu>

*Modern Campus ID (Student ID / Staff ID)
for universities in Europe & the world*



Demo → Download an eduTAP created pass



<https://demo.edutap.eu/>



 Add to Google Wallet

 Add to Apple Wallet

eduTAP – educational tapping

tap to ...

- ✓ identify yourself,
- ✓ verify your status,
- ✓ proof your entitlement,
- ✓ claim a discount,

tap to access a service on-site directly:

- ✓ tap to pay (open and close loop-payments),
- ✓ tap to open a door,
- ✓ tap to lend a book,
- ✓ tap to take a campus shuttle.

TAP TO PAY



no app, just tap!

<https://eduTAP.eu>

eduTAP – educational tapping – What is eduTAP?

- eduTAP the concept
 - Splitting integrated campus cards into **dedicated service passes** and a **harmonized identity pass**
 - Leveraging the full potential of digital wallets
 - **Use existing**, secure and proven contactless **standards** – do not invent new standards

→ Focus on **User Experience** and **interoperability**

- eduTAP the software – to enable HEIs and Service Providers to:
 - create, issue, and manage passes within the smart device wallet
 - interact with wallet passes to give access to services, identify, and verify the status of students and staff members

→ eduTAP helps to fulfill the
great vision of the European Student Card
and **enables interoperability** and a **better user experience**

We'd love to hear from you!

Get in touch:

eduTAP Project (<https://eduTAP.eu/>)

- Alexander Loechel (LMU Munich)
- Simon Lund (LMU Munich)
- José Filipe Alves (U.Porto)
- Morgan Persson (Lunds universitet)

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